GAUSSIAN BLOCK INTEGRATION AND FREE ACTION-DENSITY CORRELATIONS

Abstract. We derive Gaussian integration on prescribed linear averaging fibres, with physical measures and residual gauge factors. The constrained minimizer, fluctuation covariance and effective action compose under successive block integrations. Fixed real coordinate integrals admit complex covariance parameters with coherent determinant branches, local moment bounds and a cost determined by the selected coordinates. For the flat four-dimensional curvature we compute the separated action-density correlations and compare their lattice and continuum kernels on a uniform class of separated tests. The quadratic data are given; nonlinear background admission and interacting comparisons are separate inputs.

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1. Introduction

Block integration separates a field into the minimizer with its prescribed average and a fluctuation with zero average. At the quadratic level this is an exact Gaussian decomposition. Its normalization depends on the constraint measure and, in a gauge theory, on the coordinates used along residual gauge directions. Those factors must be retained when integrations are composed or a gauge is changed.

We first prove the finite-dimensional constrained formula and its composition law. The proof needs positivity on the constrained space, not on every direction of the ambient field space. We then identify the coarea and gauge determinants in physical coordinates. The complex integral is defined on fixed real coordinates, with a determinant logarithm continued from a positive real form. This permits fixed measurable cutoffs and gives estimates depending on the selected dimension. A real Gaussian restricted by convex coordinate tests has separate conditional and flux identities.

The last part concerns the free curvature in four dimensions. Its separated Wick contractions have an explicit positive two-point coefficient. We keep the finite-lattice contact term, the periodic zero mode and the errors from mesh, volume and coarse conditioning. Continuum cancellations are stated at their actual scope; they are not exact assertions about every finite lattice. The continuum Wick-square notation describes separated correlation limits, not a claimed ultraviolet limit of each individual random observable.

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The Gaussian block transform and its constrained shift occur in [3, §§A–B, D]; mixed-scale constraints are treated in [4, §A]. The background-field covariance is constructed in [5, §E, Theorem 3.15]. We use the averaging geometry of [1] and the specified linear inverse, coercivity and local matrix estimates of [2]. The proofs below give the finite measure identities and free-density comparisons in the conventions stated here. They do not identify an interacting measure with the free model.

In a physical application the complete constrained Hessian includes the multiplier paired with the second derivative of the constraint. Its identification and the admission of a nonlinear critical background must be supplied. Nonquadratic action, nonconstant Haar and chart factors, and interacting cutoff corrections are outside the present Gaussian calculation.

2. A Gaussian integral with a linear constraint

Let E, F be finite-dimensional real Hilbert spaces. We use the Lebesgue measure of their orthonormal coordinates, including the physical weights when the spaces are lattice fields. The adjoint for these pairings is Q^*. On a complexification we continue it by bilinear transpose. Determinants on the zero space and the mass of its Lebesgue measure are one.

For SU(2) we use traceless Hermitian Lie coordinates with real pairing $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$. The Pauli matrices are then an orthonormal real basis.

Let $Q : E \to F$ be onto and put $K = \ker Q$, with its induced inner product and inclusion $\iota_K : K \to E$. Write $J_Q = (\det QQ^*)^{1/2}$. For every $b \in F$ define the measure $d\nu_{Q,b}$ on $Q^{-1}(b)$ as J_Q^{-1} times its induced Lebesgue measure. We use $\delta_F(Qx - b)\,dx$ for this measure. Indeed the orthogonal right inverse $Q^*(QQ^*)^{-1}$ maps F to $K^\perp$ with volume factor J_Q^{-1}. This proves the coarea formula with db the declared Lebesgue measure of F. Replacing that right inverse by any other right inverse adds a shear into K and preserves the volume factor. If $S : F \to F'$ is an isomorphism, then $d\nu_{SQ,Sb} = |\det S|^{-1} d\nu_{Q,b}$, since the fibre is the same and $J_{SQ} = |\det S| J_Q$.

For a symmetric precision $T : E \to E$ put $T_K = \iota_K^* T \iota_K$. If $T_K > 0$, write

$$I(T, Q; b) = \int_E e^{-\langle x, Tx\rangle/2}\, d\nu_{Q,b}(x), \qquad I(T, Q) = I(T, Q; 0).$$

The constrained shift and the normalization below are the finite linear algebra of the block transform in [3, §D, eqs. (1.64)–(1.65)].

Theorem A (Constrained minimizer and fluctuation law)**.** Suppose that $Q : E \to F$ is onto, $T = T^*$, and $T_K > 0$. There is a unique linear map $H : F \to E$ with

$$QH = I_F, \qquad \iota_K^* T H = 0.$$

For $Qx = b$, the minimum of $\langle x, Tx\rangle/2$ is attained uniquely at Hb and equals $\langle b, \Sigma b\rangle/2$, where $\Sigma = H^*TH$. Let $C = \iota_K T_K^{-1} \iota_K^*$ and let γ_C be the probability law on E of the centred Gaussian on K with covariance T_K^{-1}. Then for each real b and bounded Borel $\varphi : E \to \mathbb{C}$,

$$\int_E \varphi(x) e^{-\langle x, Tx\rangle/2}\, d\nu_{Q,b}(x) = I(T, Q) e^{-\langle b, \Sigma b\rangle/2} \int_E \varphi(Hb + u)\, d\gamma_C(u). \tag{2.i}$$

The same identity holds for absolutely integrable φ. With $m = \dim F$, the zero-fibre constant is

$$\begin{aligned} I(T, Q) &= (2\pi)^{\dim K/2} (\det T_K)^{-1/2} J_Q^{-1} = (2\pi)^{\dim K/2} |\det \mathbb{S}(T, Q)|^{-1/2}, \\ \det \mathbb{S}(T, Q) &= (-1)^m J_Q^2 \det T_K, \qquad \mathbb{S}(T, Q) = \begin{pmatrix} T & Q^* \\ Q & 0 \end{pmatrix}. \end{aligned} \tag{2.ii}$$

■□

Proof. For a right inverse L_Q of Q, the map

$$H = L_Q - \iota_K T_K^{-1} \iota_K^* T L_Q$$

is still a right inverse and satisfies $\iota_K^* T H = 0$. A difference of two such maps takes values in K, where the vanishing equation forces that difference to be zero. Writing $x = Hb + u$, $u \in K$, removes the cross term in the precision. The remaining quadratic form in u is positive. This proves the minimizer and the effective quadratic action. Coarea gives $J_Q^{-1} du$ on the fibre in these coordinates. Diagonalization of T_K and the one-dimensional Gaussian integral give the first line of eq. 2.ii and the probability law in eq. 2.i.

Use the linear isomorphism $(u, b, \lambda) \mapsto (\iota_K u + Hb, \lambda)$ on the two spaces of the saddle. Its absolute determinant is J_Q^{-1}. The saddle form in these coordinates is the direct sum of T_K and $\begin{pmatrix} \Sigma & I_F \\ I_F & 0 \end{pmatrix}$. Exchanging the latter two block columns makes this matrix block triangular with identity diagonal and contributes the sign $(-1)^m$. This proves the determinant identity. □

The saddle is invertible and its inverse in the original physical pairings is

$$\mathbb{S}(T, Q)^{-1} = \begin{pmatrix} C & H \\ H^* & -\Sigma \end{pmatrix}. \tag{2.iii}$$

In fact $Tx + Q^*\lambda = f$, $Qx = b$ first gives $x = Cf + Hb$ by testing on K. Applying H^* to the first equation then gives $\lambda = H^* f - \Sigma b$. Adding a row quadratic term $Q^* A Q$, $A = A^*$, leaves $H, C, I(T, Q)$ unchanged and replaces Σ by $\Sigma + A$.

When $T > 0$ on all of E, set $G_0 = T^{-1}$. The equality $TH = Q^*\Sigma$ follows from stationarity and the fact that $\operatorname{ran} Q^* = K^\perp$. Applying QG_0 gives

$$\Sigma = (QG_0Q^*)^{-1}, \qquad H = G_0 Q^* \Sigma, \qquad C = G_0 - HQG_0. \tag{2.iv}$$

For the last formula the right side takes values in K and its product with $\iota_K^* T$ is ι_K^*. These two properties characterize C. A Gaussian of covariance G_0 therefore decomposes as $Hb + u$, where b has covariance Σ^{-1}, u has law γ_C, and the two variables are independent. This follows by integrating eq. 2.i in b. The constrained formula requires only the positive restriction T_K.

3. Successive constraints

Let $M : E \to F$ and $N : F \to G$ be onto. Here G is a third real Hilbert space with its declared Lebesgue measure. For $T = T^*$ positive on $\ker(NM)$, use H_M, C_M, Σ_M for the operators of theorem A with (T, M). For the second step use H_N, C_N, Σ_N with (Σ_M, N). These exist: T is positive on $\ker M$, and $H_M b$ is a nonzero vector in $\ker(NM)$ for every $b \neq 0$ in $\ker N$. Thus Σ_M is positive on $\ker N$.

Proposition 3.1 (Gaussian composition)**.** Under these hypotheses the operators for (T, NM) satisfy

$$H_{NM} = H_M H_N, \qquad C_{NM} = C_M + H_M C_N H_M^*, \qquad \Sigma_{NM} = \Sigma_N. \tag{3.i}$$

For every $c \in G$ and every bounded Borel test $\varphi : E \to \mathbb{C}$,

$$\begin{aligned} &\int_E \varphi(x) e^{-\langle x, Tx\rangle/2} \, d\nu_{NM,c}(x) \\ &\qquad = I(T, M) I(\Sigma_M, N; c) \, \mathbb{E}\big[\varphi(H_M H_N c + U + H_M V)\big], \end{aligned} \tag{3.ii}$$

where U, V are independent with laws γ_{C_M} and γ_{C_N}. In particular

$$I(T, NM) = I(T, M) I(\Sigma_M, N). \tag{3.iii}$$

■

Proof. Write $x = u + H_M(v + H_N c)$, where $u \in \ker M$ and $v \in \ker N$. With c free this is an isomorphism onto E of absolute determinant $J_M^{-1} J_N^{-1}$, by the two coarea changes. At $c = 0$ it

maps the first two slots onto $\ker(NM)$. The last slot is a right inverse to NM, so the product $J_M^{-1}J_N^{-1}\,du\,dv$ is exactly $d\nu_{NM,c}$ in these coordinates for every c. The precision in them is

$$\langle x, Tx\rangle = \langle u, Tu\rangle + \langle v, \Sigma_M v\rangle + \langle c, \Sigma_N c\rangle$$

by the two stationarity equations. Thus $u = v = 0$ is the unique minimum on each fibre and the two centred Gaussian factors are independent. Applying the single-fibre normalization to them gives eq. 3.ii. Computing the covariance of $U + H_M V$ proves the middle identity in eq. 3.i. Taking the constant test and $c = 0$ proves the product. □

For any finite chain of onto constraints, assume positivity on the kernel of their total composition. Use the composite of the remaining maps as N in proposition 3.1. The effective precision of the first step is then positive on the kernel of that remaining composite. Induction gives the successive law for all the stages. A second-stage quadratic weight on Mx can be included by adding it to Σ_M. For example, suppose that the supplied complete first-stage Hessian is B_1 and its lift H satisfies $MH = I$ and $\iota^*_{\ker M} B_1 H = 0$. Suppose that the first-stage gauge coordinate G_1 satisfies $G_1 H = 0$. If the supplied total form is

$$B_1 + G_1^* G_1 + M^*(A_2 + G_2^* G_2)M,$$

the first-step restricted precision is $(B_1 + G_1^* G_1)|_{\ker M}$ as a quadratic form. The second-step precision is $H^* B_1 H + A_2 + G_2^* G_2$. Positivity on the respective integration kernels permits eq. 3.iii. This is the algebraic composition for a prescribed gauge and Hessian splitting.

The corresponding identity for logarithms uses the same reference at both stages. If the first factor is evaluated at preceding data $p(z)$ and the second at z, define each normalized logarithm by subtracting its value at $p(z_*)$ or z_*, respectively. The total normalized logarithm is then the sum of the first evaluated at $p(z)$ and the second at z. This follows by applying eq. 3.iii at z and z_*. Any constant measure factor cancels in that difference.

3.1. **The flat curvature slices.** Fix an odd blocking factor $L \geq 3$ and a nonnegative integer K of levels. Use a periodic hierarchy in which each fine period is divisible by L^K and has at least two sites at level K, as in the admitted geometry of [1, 2]. Let $\mathcal{E} = \operatorname{ran} \mathrm{d}$ be the exact fine two-cochains on a compatible homogeneous periodic lattice, with the counting Hilbert pairing on raw plaquette fluxes. Write Q_j^{pl} and Q_j^{st} for the composed plain and straight link rows at the identity in raw link coordinates. Write d_j for the coboundary at level j, with $\mathrm{d} = \mathrm{d}_0$. Let $\mathcal{R}_j$ be the actual level-j sheet average on $\mathcal{E}$, with $\mathcal{R}_0 = I$. In raw flux units, for a coarse plaquette P based at y, this average is

$$(\mathcal{R}_j F)(P) = L^{-4j} \sum_{z \in B^j(y)} \sum_{p \in \mathrm{Sq}^j(z;P)} F(p).$$

Here $B^j(y)$ is the level-j block of L^{4j} fine sites, and $\mathrm{Sq}^j(z;P)$ is the square of L^{2j} oriented fine plaquettes parallel to P, translated to start at z. Multiplicities are retained under periodic projection. Put $W_j = \mathcal{E} \ominus \ker \mathcal{R}_j$ and denote by P_j the orthogonal projection onto W_j, extended by zero on $\mathcal{E}^\perp$. In particular $P_0 = P_{\mathrm{ex}}$, the exact-form projection. The sheet composition and deterministic projection identities are given in [2, The deterministic curvature slices]. For the hard constraints fix the internal coordinate spanning tree rooted at the center of each block, as in [2, Kernel of a flat curvature slice].

Proposition 3.2 (Independent curvature slices)**.** Let F_{free} be the centred Gaussian with covariance $g^2 P_{\mathrm{ex}}$, $g > 0$. Its conditional mean given $\mathcal{R}_j F_{\mathrm{free}}$ is $P_j F_{\mathrm{free}}$, the least-norm exact cochain with that datum. The decomposition

$$F_{\mathrm{free}} = P_K F_{\mathrm{free}} + \sum_{j=1}^{K} (P_{j-1} - P_j) F_{\mathrm{free}} \tag{3.iv}$$

has independent centred terms, with covariances $g^2 P_K$ and $g^2(P_{j-1} - P_j)$. At step j, take the intermediate link fields whose next straight block average $S_L a$ is zero and whose prescribed internal spanning-tree links are zero. Parametrize this subspace by its remaining nonpivot variables. If B_j sends these variables to their least-norm fine curvature, and $A_j = B_j^* B_j$ is the restricted unit-coupling step precision in those variables, the precision at coupling g is $g^{-2} A_j$. Its fluctuation covariance pushes forward to

$$B_j (g^2 A_j^{-1}) B_j^* = g^2 (P_{j-1} - P_j).$$

■

Proof. Use the orthogonal split $\mathcal{E} = W_j \oplus \ker \mathcal{R}_j$. The two Gaussian components are independent since the law on $\mathcal{E}$ has scalar covariance. The restriction of $\mathcal{R}_j$ to W_j is one-to-one and onto the range of $\mathcal{R}_j$, so the datum fixes the first component. The second is centred. Orthogonality gives the asserted minimum. At the identity, the plain and straight link derivatives differ by a coarse gradient; their coarse coboundaries therefore agree. The exact flux formula in [1, The flat derivative and its curl] gives, for both raw link rows,

$$\mathrm{d}_j Q_j^{\mathrm{pl}} = \mathrm{d}_j Q_j^{\mathrm{st}} = \mathcal{R}_j \mathrm{d}. \tag{3.v}$$

Thus observing either averaged link field modulo closed cochains gives the same conditioning, since the coarse coboundary identifies that quotient with the exact coarse cochains.

The ranges W_j are nested by sheet composition. The successive differences of their projections are orthogonal projections with pairwise orthogonal ranges. Their telescoping sum proves eq. 3.iv, and orthogonal coordinates of the scalar Gaussian prove the stated independence and covariances. Finally the hard-tree parametrization and flat slice result of [2, Kernel of a flat curvature slice] identify B_j as an injection onto $W_{j-1} \cap W_j^{\perp}$. For an injective map B_j, the orthogonal projection onto its range is $B_j (B_j^* B_j)^{-1} B_j^*$, as follows by testing the error against that range. This gives the lifted fluctuation covariance in the same counting units. □

Proposition 3.3 (The flat curvature law in two gauges)**.** On the compatible periodic hierarchy of proposition 3.2 at the identity connection, use the plain and straight rows and the counting inner products defined there. For a step $1 \le j \le K$ put $A = \mathcal{R}_{j-1}|_{\mathcal{E}}$ and $V = AA^*$, viewed on $\mathcal{E}_{j-1} = \operatorname{ran} \mathrm{d}_{j-1}$. The stored real cochains carry counting inner products. Their linear subspaces carry the induced inner products and Lebesgue measures, which also determine all adjoints. Use the specified flat quadratic energy

$$\frac{1}{2g^2} \langle \mathrm{d}_{j-1} a, V^{-1} \mathrm{d}_{j-1} a \rangle$$

on the intermediate links. Constrain the next plain row to zero and add the soft Gaussian penalty $\langle G_{\mathrm{con}} a, W_{\mathrm{con}} G_{\mathrm{con}} a \rangle / 2$, where W_{con} is any fixed positive definite matrix on the contour coordinates. Here $G_{\mathrm{con}} a$ is the arithmetic mean of the contour path sums from the block center to each noncenter site. Alternatively, constrain the next straight row to zero and set the internal spanning-tree links to zero in each block. The two normalized laws have the same curvature pushforward. It has precision form $g^{-2} V^{-1}$ restricted to the exact intermediate curvatures annihilated by the next plaquette average. Its least-norm lift to the bare lattice has covariance $g^2(P_{j-1} - P_j)$. ■

Proof. Use one colour. Both one-step link rows are onto. A right inverse assigns the datum of each positive coarse bond to all fine bonds crossing its literal outgoing block face on the periodic lift, and sets internal bonds to zero. Every translated straight segment crosses that face once and every internal contour sum is zero. This construction works on the stored bond occurrences, including across the periodic seam. Composing and taking curls shows that A is onto $\mathcal{E}_{j-1}$. The least-norm lift of an intermediate curvature h is $A^*V^{-1}h$. Its squared norm is $\langle h, V^{-1}h\rangle$. This is the flat effective curl energy used in [2, Proposition 4.1 and Corollary 4.3].

Write q_{pl} for evaluation on block centers and q_{st} for the arithmetic mean on each block. On a site gradient $\mathrm{d}_{j-1}\lambda$ the next raw link rows give, respectively, $\mathrm{d}_j q_{\mathrm{pl}}\lambda$ and $\mathrm{d}_j q_{\mathrm{st}}\lambda$. The first identity follows by telescoping each full contour from center to center. The second follows by telescoping the straight segments and averaging their endpoints. Both site rows are onto. Both link rows send a constant directional field c to Lc. On either periodic lattice a closed link field is a gradient plus a constant directional field. For instance, at each nonzero Fourier frequency the coboundary symbol is wedge multiplication by the nonzero difference vector, whose kernel on links is its span. The zero frequency is the constant directional part.

Let h be exact with zero next plaquette average and choose any a with $\mathrm{d}_{j-1}a = h$. Its next link datum is closed by eq. 3.v for a single step. Subtract a constant fine field to remove the constant part of that datum, then subtract the gradient of a site lift to remove its gradient part. This works for either row. Thus the curl maps both zero-datum spaces onto the same curvature subspace. Its null space for either row consists of gradients with $q_{\mathrm{pl}}\lambda = 0$ or $q_{\mathrm{st}}\lambda = 0$, respectively. Indeed the constant part of a closed field in the zero-datum space must vanish, and a constant site value can be subtracted from λ to impose the indicated condition.

On the plain null space $G_{\mathrm{con}}\mathrm{d}_{j-1}\lambda$ lists the noncenter values of λ, since the contour sum is $\lambda(x) - \lambda(y)$ and $\lambda(y) = 0$ at the center y. It is a bijection. Choose a linear lift of h to the plain zero-datum space and add the null-space gradient. This linear splitting has a constant Jacobian, and the row coarea factor J_Q^{-1} is constant as well. Changing the residual variables to $G_{\mathrm{con}}a$ in the Gaussian integral has constant Jacobian, and translation by the contour values of the lift leaves the integral unchanged. The soft integral hence multiplies the curvature density by a constant independent of h.

For the straight row, integrate the desired tree link values along each internal tree to obtain a site function. Subtract its block means and then subtract its gradient from a. The row stays zero, and the resulting tree links vanish. A residual gradient vanishing on the tree has a constant potential in each block, and its zero block means force it to vanish. The hard-tree space thus maps bijectively to the same curvature subspace with constant Jacobian. In both cases normalization cancels the constant, giving the stated restriction of the curvature precision. The least-norm lift maps that subspace onto $W_{j-1} \cap W_j^{\perp}$ by sheet composition. Its energy is the bare squared norm divided by $2g^2$, which proves the last covariance identity. □

For the background current of the unit-coupling Wilson quadratic form and $F \in \mathcal{E}$, write $P_j F = (\mathcal{R}_j|_{\mathcal{E}})^*\phi$ for some coarse plaquette field ϕ. The restricted adjoint includes P_{ex}, and $\mathrm{d}^* P_{\mathrm{ex}} = \mathrm{d}^*$ because d takes values in $\mathcal{E}$. Taking the transpose of eq. 3.v gives

$$\mathrm{d}^* P_j F = (Q_j^{\mathrm{st}})^* \mathrm{d}_j^* \phi. \tag{3.vi}$$

Thus the current has the row-multiplier form even in the flat quadratic theory. This is also the deterministic identity in [2, The deterministic curvature slices]. Multiplying the quadratic action by g^{-2} multiplies this current by the same factor.

4. Measure factors and changes of gauge

The Gaussian normalizer depends on the measures of both the field and constraint spaces. A gauge change preserves the physical quadratic form but changes the coordinates integrated along its null directions. We retain these two effects separately.

4.1. **Physical coordinates.** All determinants in this section use orthonormal coordinates in the declared real Hilbert spaces. Suppose that a stored coordinate array on E has inner product $x^{\mathsf{t}}W_E y$, with $W_E > 0$, and that F has Gram matrix $W_F > 0$. Then

$$dx_E = (\det W_E)^{1/2}\,dx, \qquad \delta_F(y) = (\det W_F)^{-1/2}\delta_{\mathrm{array}}(y), \qquad Q^* = W_E^{-1}Q^{\mathsf{t}}W_F. \tag{4.i}$$

Here t is the ordinary array transpose. The exponent of the operator T is $x^{\mathsf{t}}W_E Tx/2$. For the already defined $I(T,Q)$,

$$I(T,Q) = \left(\frac{\det W_E}{\det W_F}\right)^{1/2} \int_{\mathbb{R}^{\dim E}} \delta_{\mathrm{array}}(Qx)e^{-x^{\mathsf{t}}W_E Tx/2}\,dx. \tag{4.ii}$$

Indeed $W_E^{1/2}$ and $W_F^{1/2}$ carry the two physical inner products to Euclidean ones. Their Jacobians give the first two formulas, and the defining identity for an adjoint gives the third. These changes include every lattice-volume weight. Replacing Q^* by Q^{t} before making this change would alter both the covariance and the coarea factor.

4.2. **A relative gauge normalizer.** Let $E, F, \mathcal{H}, \mathcal{K}$ be finite-dimensional real Hilbert spaces. Assume injective maps $D : \mathcal{H} \to E$ and $d : \mathcal{K} \to F$, onto maps $Q : E \to F$ and $R : \mathcal{H} \to \mathcal{K}$, and the identity

$$QD = dR. \tag{4.iii}$$

Let $B = B^*$, $BD = 0$, and suppose that B is nonnegative on $\ker Q$, with null space there exactly $D(\ker R)$. Put $\mathcal{H}_0 = \ker R$ and $D_0 = D|_{\mathcal{H}_0}$. A residual gauge coordinate is a map $G : E \to Z$, where Z is a real Hilbert space with a fixed norm and measure, $\dim Z = \dim \mathcal{H}_0$, and $GD_0 : \mathcal{H}_0 \to Z$ is invertible. Its determinant uses the induced Hilbert measure of $\mathcal{H}_0$ and the declared measure of Z. The operator $B + G^*G$ is positive on $\ker Q$: a zero vector for its quadratic form must equal $D_0 v$, and $GD_0 v = 0$ then gives $v = 0$.

Theorem B (Relative gauge normalization)**.** Under the preceding hypotheses, the quantity

$$\mathcal{J}(B,Q,R,G) = \sqrt{\det(RR^*)}\,|\det(GD_0)|\,I(B + G^*G, Q) \tag{4.iv}$$

does not depend on the residual coordinate G. Let $S : E \to \mathcal{K}$, and suppose

$$Q' = Q + dS, \qquad R' = R + SD$$

are onto. If the same hypotheses hold for Q', R' and an admitted residual coordinate G', then

$$\mathcal{J}(B,Q',R',G') = \mathcal{J}(B,Q,R,G).$$

Only the coarse gauge component of the constraint changes. ■□

Proof. We express the normalizer on the fixed physical quotient. Use the orthogonal complements $E_\perp = (\operatorname{ran} D)^\perp$ and $F_\perp = (\operatorname{ran} d)^\perp$, and let P be the orthogonal projection onto $F_\perp$. Define $\overline{Q} = PQ|_{E_\perp}$ and $B_\perp = B|_{E_\perp}$ as a quadratic form. The map $\overline{Q}$ is onto: for a vector of $F_\perp$, choose a preimage under Q and discard its $\operatorname{ran} D$ component, which changes Q only in $\operatorname{ran} d$. The restriction of $B_\perp$ to $\ker \overline{Q}$ is positive. In fact $Qu = d\eta$ there, and an h with $Rh = \eta$ gives $u - Dh \in \ker Q$. Its action is that of u. If it is zero, $u - Dh = D_0 v$, whence u lies in $\operatorname{ran} D \cap E_\perp = \{0\}$.

Write $a = u + Dv$, with $u \in E_\perp$, $v \in \mathcal{H}$. Its Jacobian is $J_D = \sqrt{\det(D^*D)}$. Set $L = (d^*d)^{-1}d^*Q|_{E_\perp}$. Splitting F into $F_\perp \oplus \operatorname{ran} d$ gives

$$Qa = \overline{Q}u + d(Lu + Rv), \qquad \delta_F(Qa) = J_d^{-1}\delta_{F_\perp}(\overline{Q}u)\delta_{\mathcal{K}}(Lu + Rv), \qquad J_d = \sqrt{\det(d^*d)}.$$

The coarea formula for R now writes $v = v_0 - R^*(RR^*)^{-1}Lu$, $v_0 \in \mathcal{H}_0$, with factor $J_R^{-1} = (\det RR^*)^{-1/2}$. Since $BD = 0$, the action is $\langle u, Bu\rangle/2$. The gauge integral is

$$\int_{\mathcal{H}_0} \exp\{-\tfrac{1}{2}|G(u - DR^*(RR^*)^{-1}Lu) + GD_0v_0|^2\}\, dv_0 = \frac{(2\pi)^{(\dim\mathcal{H}-\dim\mathcal{K})/2}}{|\det(GD_0)|}.$$

Combining these exact changes of variables proves

$$\mathcal{J}(B, Q, R, G) = \frac{J_D}{J_d}(2\pi)^{(\dim\mathcal{H}-\dim\mathcal{K})/2} I(B_\perp, \overline{Q}). \tag{4.v}$$

Every integral is finite by the established quotient positivity. The right side is independent of G. It is also unchanged by $Q \mapsto Q + dS$, because $Pd = 0$; all its other data are fixed. This proves both conclusions. □

Corollary 4.1 (A residual gauge with a stabilizer)**.** In theorem B, allow D, d to have kernels. Require $D|_{\ker R}$ to be injective and keep the other hypotheses, including that the null space of B on $\ker Q$ is $D(\ker R)$. Then the same invariant $\mathcal{J}(B, Q, R, G)$, using the original Hilbert measures in eq. 4.iv, is independent of G and is unchanged by the stated change of constraint. Require the corresponding hypotheses for the new R', Q', G'. ■□

Proof. Write $\mathcal{H}_s = \ker D$, $\mathcal{K}_s = \ker d$, and let $\mathcal{H}_e, \mathcal{K}_e$ be their orthogonal complements. The restriction $A = R|_{\mathcal{H}_s}$ takes values in $\mathcal{K}_s$ by the gauge-chain identity. It is injective by injectivity of $D|_{\ker R}$. If $\zeta \in \mathcal{K}_s$, choose v with $Rv = \zeta$. Then $Dv \in \ker Q$ and $BDv = 0$. By the prescribed null space there is $v_0 \in \ker R$ with $Dv = Dv_0$. Thus $v - v_0 \in \mathcal{H}_s$ and $A(v - v_0) = \zeta$. This proves that A is an isomorphism.

In the orthogonal decompositions of the two site spaces, write

$$R = \begin{pmatrix} A & B_R \\ 0 & C_R \end{pmatrix}.$$

The map $C_R : \mathcal{H}_e \to \mathcal{K}_e$ is onto. The effective gradients $D_e = D|_{\mathcal{H}_e}$ and $d_e = d|_{\mathcal{K}_e}$ are injective and satisfy $QD_e = d_eC_R$. The original $\ker R$ is the graph $w \mapsto (-A^{-1}B_Rw, w)$ over $\ker C_R$. Let $\pi : \ker R \to \ker C_R$ be the projection onto its second component, with absolute determinant J_π in the induced Hilbert measures. In particular $D|_{\ker R} = (D_e|_{\ker C_R})\pi$. The effective chain therefore has the same residual gauge image and the same required nondegeneracy on $\ker Q$.

The two coarea factors satisfy

$$J_RJ_\pi = |\det A|J_{C_R}. \tag{4.vi}$$

To verify this, use $w \in \ker C_R$ and $r = (r_s, r_e) \in \mathcal{K}$ as coordinates on $\mathcal{H}$. Set

$$v_e = w + L_Cr_e, \qquad v_s = A^{-1}(r_s - B_Rv_e),$$

where L_C is the orthogonal right inverse of C_R. Then $R(v_s, v_e) = r$, and the triangular Jacobian of this change is $|\det A|^{-1}J_{C_R}^{-1}$. Alternatively, first use the coarea factor J_R^{-1} with the induced measure on $\ker R$, and then use J_π^{-1} to express that measure in w. Equality of these Jacobians proves eq. 4.vi.

Because $|\det(GD_0)| = |\det(GD_e|_{\ker C_R})|J_\pi$, the original invariant equals $|\det A|$ times the invariant for the effective chain, with its induced orthogonal Hilbert measures. Theorem B applies to the effective chain. Under $R' = R + SD$ the map A is unchanged, and the effective map is $C'_R = C_R + P_{\mathcal{K}_e}SD_e$. This is exactly the same constraint change for that chain because d annihilates $\mathcal{K}_s$. The effective invariance and the fixed factor $|\det A|$ prove both claims with the original measures. □

The factor involving R cannot be dropped. For example, take $E = \mathbb{R}^2$, $F = \mathcal{H} = \mathcal{K} = \mathbb{R}$, $Dv = (0, v)$, $d = 1$, $Q(u, v) = \alpha u + \beta v$ with $\beta \neq 0$, and $B = \mathrm{diag}(\lambda, 0)$, $\lambda > 0$. There is no residual

gauge coordinate. Then $R = \beta$ and

$$I(B, Q) = |\beta|^{-1}\sqrt{2\pi/\lambda}, \qquad \mathcal{J}(B, Q, R, 0) = \sqrt{2\pi/\lambda}.$$

Changing the coarse gauge constraint changes the uncorrected integral even though it leaves the physical quotient unchanged.

4.3. **The background gauge and stabilizers.** Set $\Delta = D^*D$, $P_0 = I - R^*(RR^*)^{-1}R$, and use $G = P_0D^* : E \to \mathcal{H}_0$. Then $GD_0 = \Delta_{00}$, the compression of Δ to $\mathcal{H}_0$, and $G^*G = DP_0D^*$. The saddle determinant formula for $\mathbb{S}(A, Q) = \left(\begin{smallmatrix} A & Q^* \\ Q & 0 \end{smallmatrix}\right)$ therefore gives

$$\mathcal{J} = \frac{|\det \mathbb{S}(\Delta, R)|}{\sqrt{\det(RR^*)}} I(B + DP_0D^*, Q). \tag{4.vii}$$

Indeed $|\det \mathbb{S}(\Delta, R)| = \det(RR^*) \det \Delta_{00}$; substitution into eq. 4.iv proves the formula. When $RR^* = I$, its logarithm is

$$\log \mathcal{J} = -\tfrac{1}{2} \log |\det \mathbb{S}(B + D(I - R^*R)D^*, Q)| + \log |\det \mathbb{S}(\Delta, R)| + \tfrac{\dim E - \dim F}{2} \log(2\pi). \tag{4.viii}$$

The second determinant is a constrained site determinant. It is not an arbitrary determinant of a gauge operator.

For complex data all transposes are bilinear continuations of the physical real adjoints. For a saddle with target dimension m, the real positive factor is $(-1)^m \det \mathbb{S}(T, Q)$. Continue its logarithm and square root from that positive value on the admitted branch domain. The site saddle uses its own target dimension $\dim \mathcal{K}$. Absolute values are not holomorphic normalizations.

4.4. **The specified physical fibre.** We now state the use of the preceding algebra on lattice data. The geometric maps and their derivatives are those of [1]; the nondegenerate quadratic forms are the inputs treated in [2]. No existence of a nonlinear critical background is inferred from the following Gaussian identity.

Consider a homogeneous blocking with fine mesh η and coarse mesh one. Give fine site coordinates weight η^4 and coarse site coordinates weight one. Let R_0 evaluate a site field at the distinct coarse nodes. Its physical adjoint inserts η^{-4} times each node value, so

$$R_0R_0^* = \eta^{-4} I. \tag{4.ix}$$

For an assigned block with N fine sites, the relative transported average has the form $R\lambda = N^{-1} \sum_x O_x \lambda_x$, where every O_x is a real adjoint transport. The fine-to-coarse measure ratio is N^{-1}. Therefore its physical adjoint has components O_x^*, and

$$RR^* = N^{-1} \sum_x O_x O_x^* = I. \tag{4.x}$$

Different blocks are disjoint. A homogeneous composition of these coisometries is again a coisometry, by applying $R_1R_1^* = I$ successively from the inside. The maps here are relative derivatives centered at the given real background. A derivative of a fixed-reference nonlinear chart at a nonzero coordinate is a different map.

Let T_{yx} be the mean contour transport from a parent node y to a site x at the real reference background. Endpoint covariance gives the derivative of its relative left logarithm as $\lambda(y) - \mathrm{Ad}(T_{yx})\lambda(x)$, up to the fixed overall sign. At a mean-contour axial representative $T_{yx} = I$. A representative that fixes only a single comb need not have this property. Order noncoarse sites by the hierarchy, putting a parent before its descendants, and set the final coarse-node values to zero. The resulting map on these site values is triangular, with diagonal blocks $\pm \mathrm{Ad}(T_{yx})$. These adjoint transports have determinant one. The same is true of any transported colour frames. Consequently the actual axial coordinate G_{ax} satisfies

$$\sqrt{\det(R_0R_0^*)}\, |\det(G_{\mathrm{ax}} D|_{\ker R_0})| = c_{\eta,j}, \tag{4.xi}$$

where $c_{\eta,j} > 0$ depends on the meshes, hierarchy, coordinate measures and fixed gauge weights, but not on the background. The determinant one calculation is in unweighted site coordinates; conversion by eq. 4.i accounts for its physical measure factors.

The linear coarea factor also equals the pivot-elimination factor in a local fibre chart. To see this in orthonormal coordinates, split $x = (u, p)$ so that $Q = (Q_u, Q_p)$ with Q_p invertible. The kernel is the graph $p = -Q_p^{-1}Q_u u$. Its induced measure is $\det(I + A^*A)^{1/2}du$, where $A = -Q_p^{-1}Q_u$. Since

$$QQ^* = Q_p(I + AA^*)Q_p^*, \qquad \det(I + AA^*) = \det(I + A^*A),$$

the kernel measure divided by $\sqrt{\det QQ^*}$ is $|\det Q_p|^{-1}du$. The determinant identity follows by eliminating either diagonal block of $\left(\begin{smallmatrix} I & -A^* \\ A & I \end{smallmatrix}\right)$. This is precisely the derivative of constraint elimination. In the product Haar disintegration of [1, Proposition "Local disintegration of product Haar measure"], the left exponential density at the origin is field independent. Its central value and the fixed coordinate scales multiply this linear integral by constants; its nonconstant part is a nonquadratic correction and remains outside the present calculation.

For clarity, these central Haar constants can be written explicitly in the present colour convention. In a Pauli basis of norm one, normalized SU(2) Haar measure in $U = \exp(ix \cdot \sigma)$, $|x| < \pi$, is

$$\frac{1}{2\pi^2}\left(\frac{\sin|x|}{|x|}\right)^2 d^3x.$$

Indeed SU(2) is the unit three-sphere $(\cos r, n \sin r)$. Multiplication by a unit quaternion is orthogonal, so normalized spherical volume is Haar measure. Its polar volume is $\sin^2 r\, dr\, d\Omega$, of total volume $2\pi^2$. Euclidean polar measure is $r^2 dr\, d\Omega$. Thus the coordinate $U = e^{ihA}U_0$ has central Haar density $(2\pi^2)^{-1}h^3 d^3A$. The physical bond Lebesgue measure of the pairing $h^4\langle A, A\rangle$ is $h^6 d^3A$, giving the central ratio

$$c_h = (2\pi^2)^{-1}h^{-3}. \tag{4.xii}$$

Before any gauge insertion, the linear fibre density relative to prescribed target product Haar measure therefore carries $\prod_{b\in E_f} c_{h_b} / \prod_{c\in E_c} c_{h_c}$, multiplying the physical delta integral. The sets E_f, E_c here are all fine and coarse link variables of that disintegration, not only nonpivot variables. The pivot determinant is already in the coarea factor. A subsequent rescaling of the free integration coordinates contributes its own Jacobian. These constants are retained for an absolute normalizer and cancel only in a comparison with the same coordinate and measure conventions.

Proposition 4.2 (Normalized physical Gaussian)**.** Fix the spaces, meshes, coordinate measures and soft-gauge normalization conventions. Suppose the following finite data are given at each real background parameter V, including a reference V_*. The plain and straightened constraints satisfy

$$Q = Q_0 + dS, \qquad R = R_0 + SD, \qquad Q_0 D = dR_0,$$

and the hypotheses of the relative gauge identity, with ineffective gauge directions treated by corollary 4.1, hold for both constraints. Their physical quadratic form is represented by a common symmetric B with $BD = 0$. A specified complete fibre Hessian B_{fib} obeys

$$B = B_{\mathrm{fib}} - Q^* H_c Q \tag{4.xiii}$$

for a symmetric coarse form H_c. Assume eq. 4.x and eq. 4.xi, and use the physical coarea and central Haar conventions just described. Let $\Psi(V)$ be the logarithm of the plain-fibre Gaussian normalizer minus the same logarithm at V_*. Then

$$\begin{aligned}\Psi(V) = \; & -\tfrac{1}{2}\log\left|\det \mathbb{S}(B_{\mathrm{fib}} + D(I - R^*R)D^*, Q)\right| + \log\left|\det \mathbb{S}(D^*D, R)\right| \\ & - (\text{the same two terms at } V_*)\,.\end{aligned} \tag{4.xiv}$$

In a nonlinear application the identification $B_{\mathrm{fib}} = A'' - \langle \mu, q''\rangle$ and the common form in eq. 4.xiii are supplied data. This proposition does not prove their admission or an extension to independent currents. ∎

Proof. Write c_{Haar} for the product of the declared central Haar factors, divided by the target Haar factors when that is the target convention. Write $c_{\mathrm{soft}} > 0$ for the product of any prescribed fixed soft-gauge prefactors; it is one when the quadratic gauge weights have no prefactors. Both constants are independent of V. The absolute plain-fibre normalizer and the relative invariant satisfy separately

$$Z_{\mathrm{plain}} = c_{\mathrm{Haar}} c_{\mathrm{soft}} I(B + G_{\mathrm{ax}}^* G_{\mathrm{ax}}, Q_0), \qquad \mathcal{J} = c_{\eta,j} I(B + G_{\mathrm{ax}}^* G_{\mathrm{ax}}, Q_0).$$

Thus $Z_{\mathrm{plain}} = (c_{\mathrm{Haar}} c_{\mathrm{soft}} / c_{\eta,j}) \mathcal{J}$. Apply gauge and constraint invariance, then use the background gauge on the straightened side. Formula eq. 4.viii gives the two saddle logarithms and the fixed power of 2π. The row-penalty invariance of the saddle determinant removes $Q^* H_c Q$ from the first quadratic form. Every removed constant is independent of V, so subtraction at V_* proves eq. 4.xiv. □

The second saddle matrix stays invertible when D has global zero modes, provided it is injective on $\ker R$: its compressed site form is then positive, and R is onto. Thus the formula uses finite constrained determinants; an uncanceled whole-space site determinant would not have the same property. For successive integrations all measure constants must first be retained in the exact composition formula. Cancellation in a normalized logarithm uses the same reference and the same coordinate conventions at the corresponding stage.

4.5. **Scaling the action while keeping the gauge weight.** Let V be a real d-dimensional Hilbert space and suppose

$$V = V_{\mathrm{p}} \oplus V_{\mathrm{g}}, \qquad \dim V_{\mathrm{g}} = r.$$

This is a direct sum, not necessarily orthogonal. Let K_A, K_G be symmetric quadratic forms such that K_A vanishes on V_{g}, K_G vanishes on V_{p}, both have zero mixed terms, and each is positive on its other summand. Thus $K = K_A + K_G > 0$. These are the precise splitting properties supplied by the complete fibre Hessian and residual gauge coordinate in the linear estimates; a partial Wilson Hessian need not have them.

Proposition 4.3 (The two action-weight powers)**.** For $\operatorname{Re}\tau > 0$, put $K_\tau = \tau K_A + K_G$, and continue powers from $\tau = 1$. Then

$$\det K_\tau = \tau^{d-r} \det K, \tag{4.xv}$$

$$\int_V e^{-u^{\mathsf{T}} K_\tau u/2}\, du = \tau^{-(d-r)/2} \int_V e^{-u^{\mathsf{T}} K u/2}\, du, \tag{4.xvi}$$

$$\int_V e^{(\tau-1) u^{\mathsf{T}} K_G u/2} d\mu_{(\tau K)^{-1}}(u) = \tau^{r/2}. \tag{4.xvii}$$

The last measure is the normalized Gaussian on all of V, before any fluctuation test is inserted. ■

Proof. Choose fixed bases of the two summands. The two forms are block diagonal in their concatenation, so multiplication of the first block by τ multiplies the determinant by τ^{d-r}. The square of the basis-change Jacobian is the same in numerator and denominator; orthogonality is unnecessary. Integrating the two blocks for real $\tau > 0$ proves eq. 4.xvi. For complex τ the real part remains positive on compact subsets of the right half-plane, so Gaussian domination and holomorphic continuation prove the same identity there. Finally the product of the exponential in eq. 4.xvii and the density with precision τK has exponent $-u^{\mathsf{T}} K_\tau u/2$. Its normalizing factor differs from that for K by $\tau^{d/2}$. Combining this with eq. 4.xvi gives $\tau^{r/2}$. □

Thus scaling the full Gaussian precision and compensating the gauge weight contributes separately $-d \log \tau/2$ and $r \log \tau/2$ to the logarithm of the unnormalized integral. Their sum is $-(d-r) \log \tau/2$. This identity contains no estimate for a sharply truncated or interacting Gaussian integral.

Proposition 4.4 (Successive residual gauge factors)**.** Let $R_i : \mathcal{H}_{i-1} \to \mathcal{H}_i$, $i = 1, 2$, be onto maps of finite real site spaces with fixed Hilbert measures. Let $D_i : \mathcal{H}_i \to E_i$, $i = 0, 1, 2$, be the gauge tangents of the three field spaces. Let $M : E_0 \to E_1$ and $N : E_1 \to E_2$ be onto field rows satisfying $MD_0 = D_1R_1$ and $ND_1 = D_2R_2$. For $i = 1, 2$ let Z_i be fixed real Hilbert target spaces and let $G_i : E_{i-1} \to Z_i$ be such that $G_iD_{i-1}|_{\ker R_i} : \ker R_i \to Z_i$ is invertible, using the induced site measures on these kernels. Put $G_{\mathrm{tot}} = (G_1, G_2M) : E_0 \to Z_1 \oplus Z_2$, with the orthogonal product measure on the target. Then

$$\begin{aligned} J_{R_2R_1} |\det(G_{\mathrm{tot}}D_0|_{\ker(R_2R_1)})| = J_{R_1} |\det(G_1D_0|_{\ker R_1})| \\ \cdot J_{R_2} |\det(G_2D_1|_{\ker R_2})|. \end{aligned} \tag{4.xviii}$$

■

Proof. For a right inverse L_1 of R_1, the map $\Phi(u, v) = u + L_1v$ is an isomorphism from $\ker R_1 \oplus \ker R_2$ to $\ker(R_2R_1)$. In the declared Hilbert measures the two-step coarea change has factor $J_{R_1}^{-1}J_{R_2}^{-1}$. On the other hand first passing through the composite fibre gives factor $J_{R_2R_1}^{-1} |\det \Phi|$. Thus

$$|\det \Phi| = \frac{J_{R_2R_1}}{J_{R_1}J_{R_2}}.$$

The lower block of $G_{\mathrm{tot}}D_0\Phi(u, v)$ is G_2D_1v by the first gauge-chain identity. Its upper block is $G_1D_0u + G_1D_0L_1v$. The map is triangular with the two residual gauge maps on its diagonal. Taking absolute determinants gives eq. 4.xviii. □

The number of residual site coordinates is additive across the two stages. Thus any powers of 2π used to normalize the residual gauge Gaussians also multiply. Suppose the two stages and the total form satisfy the hypotheses of theorem B or corollary 4.1. If their gauge-fixed precisions also obey the hierarchical splitting of Section 3, multiplication of the two relative gauge normalizers follows from eq. 3.iii and eq. 4.xviii, with the stated measures at the intermediate field and site spaces.

For a full periodic one-step SU(2) fibre with n coarse sites and blocking factor L, the admitted geometry has $d = 12(L^4 - 1)n$ free link coordinates and $r = 3(L^4 - 1)n$ residual site coordinates. There are $4L^4n$ stored positive fine bonds and $4n$ coarse bond constraints, each with three colours; fixing the coarse nodes leaves $(L^4 - 1)n$ site gauges. Thus $d - r = 9(L^4 - 1)n$. These counts apply to the full periodic fibre. A cut or different gauge must use its actual dimensions and splitting.

5. Complex Gaussian integrals

Use orthonormal real coordinates on the selected fluctuation space. The integrals below are over this fixed real coordinate space, with Lebesgue measure du. Matrix transposes are bilinear. Norms are Euclidean operator norms, and real-part inequalities in a Gaussian exponent are tested on real integration variables.

Theorem C (The relative covariance sector)**.** Let D_0 be a real positive definite matrix on $\mathbb{R}^d$ with $\|D_0\| \le K$, where $K > 0$, and let $D = D^{\mathsf{T}}$ satisfy

$$E = D_0^{-1/2}(D - D_0)D_0^{-1/2}, \qquad r = \|E\| < \tfrac{1}{4}.$$

Use the branch

$$\log\det D = \log\det D_0 + \operatorname{Tr}\log(I + E), \qquad d\mu_D(u) = (2\pi)^{-d/2} e^{-\frac{1}{2}\log\det D} e^{-u^{\mathsf{T}}D^{-1}u/2}\, du. \tag{5.i}$$

Then $\int d\mu_D = 1$. For $0 \le \rho \le (8K)^{-1}$,

$$\int e^{\rho|u|^2} |d\mu_D(u)| \le \exp\{C(r + K\rho)d\}, \qquad |\log\det D - \log\det D_0| \le 2dr. \tag{5.ii}$$

Here C is absolute. The exponential transform for $\ell \in \mathbb{C}^d$ is

$$\int e^{\ell^{\mathsf{T}} u} d\mu_D(u) = e^{\ell^{\mathsf{T}} D \ell/2}. \tag{5.iii}$$

In particular all odd moments of linear forms vanish. For $2m$ linear forms their moment is the sum over pairings of their covariances $\ell_i^{\mathsf{T}} D \ell_j$, with absolute value at most $(2m-1)!!\|D\|^m \prod_{i=1}^{2m} \|\ell_i\|$. The absolute moment for $j \geq 1$ satisfies

$$\int \prod_{i=1}^{j} |\ell_i^{\mathsf{T}} u| \, |d\mu_D(u)| \leq e^{Crd} (C\sqrt{Kj})^j \prod_{i=1}^{j} \|\ell_i\|. \tag{5.iv}$$

The measure on the zero-dimensional space has mass one. ■□

Proof. The logarithm in eq. 5.i is the norm-convergent series for $\log(I+E)$. Its trace gives the branch of the scalar determinant logarithm because $\det \exp B = \exp(\operatorname{Tr} B)$ for a complex square matrix B. The series bounds the trace difference by $d \sum_{k \geq 1} r^k/k \leq 2dr$.

Make the real change $u = D_0^{1/2} v$ and write

$$R_\rho = \operatorname{Re}(I+E)^{-1} - 2\rho D_0.$$

The real part here is entrywise. The matrix is real symmetric, and the inverse series gives $\|R_\rho - I\| \leq r/(1-r) + 2\rho K \leq 7/12$. Thus $R_\rho > 0$. Real Gaussian integration gives the exact identity

$$\int e^{\rho|u|^2} |d\mu_D(u)| = |\det(I+E)|^{-1/2} (\det R_\rho)^{-1/2}. \tag{5.v}$$

The singular values of $I+E$ are at least $1-r$. Every eigenvalue of R_ρ is at least $1 - r/(1-r) - 2\rho K$. Using $-\log(1-x) \leq Cx$ for $0 \leq x \leq 7/12$ proves the first bound in eq. 5.ii.

For fixed D_0, the density and its exponential transform are holomorphic in the independent complex symmetric entries of E on the open norm ball. Indeed on each compact set there is an integrable Gaussian majorant, also after multiplying by $e^{M|u|}$ for any fixed M. The normalization and eq. 5.iii hold on the real symmetric ball by diagonalization and the real one-dimensional Gaussian integral. For the exponential transform, the real identity first continues in ℓ, using its entire dependence. The real symmetric ball has a full set of real coordinates for the complex symmetric space. The one-variable identity theorem in each coordinate on a small product of discs, followed by connectedness of the norm ball, proves both identities throughout the sector.

Differentiate eq. 5.iii in the scalar coefficients of a finite list of linear forms. The exponential of the quadratic form gives precisely the pairings. For the absolute moment, the absolute density is its total mass times the real Gaussian of covariance $D_0^{1/2} R_0^{-1} D_0^{1/2}$, by eq. 5.v with $\rho = 0$. Its covariance norm is at most $3K/2$. A real linear functional of this Gaussian has L^j-norm at most $C\sqrt{Kj}$ times its coefficient norm. This follows from the one-dimensional Gaussian even moments, using the next even integer if necessary. Splitting a complex linear functional into its real and imaginary parts gives the same bound with a fixed factor. Apply Hölder's inequality with exponent j to each of the j factors and use eq. 5.ii. □

Corollary 5.1 (Parameters and localization cost)**.** Let $D(s)$ be holomorphic on a finite parameter domain and satisfy the sector of theorem C about a fixed real comparison matrix D_0. Suppose $F_s(u)$ is Borel in the real variable u, holomorphic in s for each u, and $|F_s(u)| \leq A e^{\rho|u|^2}$ on each closed parameter polydisc under consideration, with $0 \leq \rho \leq (8K)^{-1}$. A fixed bounded Borel test, including a characteristic function, is allowed. Then $I(s) = \int F_s \, d\mu_{D(s)}$ is holomorphic. If the closed polydisc of radii $t_\Delta > 0$ centered at s is in the given domain and r_* bounds $\|E\|$ on it, then

$$|\partial_s^\nu I(s)| \leq A \, \nu! \prod_{\Delta : \nu_\Delta > 0} t_\Delta^{-\nu_\Delta} \exp\{C(r_* + K\rho)d\}. \tag{5.vi}$$

In this estimate it suffices to vary the coordinates with $\nu_\Delta > 0$. The same Cauchy bound applies to a finite set of directional derivatives in the given field and current norms, using only a polydisc of increments contained in the original parameter domain. For the relative log determinant, the same derivative bound uses $2dr_*$ in place of the expectation bound.

If each of N localization cubes carries at most aM^4 selected real coordinates, the exponential cost in eq. 5.vi is at most $\exp\{Ca(r_* + K\rho)M^4N\}$. The corresponding determinant difference costs at most $2ar_*M^4N$ in its logarithm. ■□

Proof. The lower bound on R_ρ in the proof of the theorem supplies an integrable majorant uniform on the chosen polydisc. Joint measurability follows by approximating the parameter by finite grids and using its pointwise continuity. Continuity and Morera's theorem, with Fubini on one coordinate disc at a time, prove holomorphy. Cauchy's formula in the differentiated variables proves eq. 5.vi. For the directional version compose with the finite affine map from the coordinate discs to the given domain. For the determinant apply the same formula to the trace-log bound in eq. 5.ii. The count is $d \le aM^4N$, with each selected coordinate counted once. □

On independent cube discs of radius $\mathcal{S} > 1$, the difference between $s_\Delta = 1$ and $s_\Delta = 0$ costs at most $(R-1)^{-1}$ times the supremum on $|s_\Delta| = R$, for $1 < R < \mathcal{S}$. Indeed the difference of the two Cauchy kernels is $1/[z(z-1)]$. Taking differences in distinct coordinates multiplies these factors. This gives the same cost per cube for mixed first differences of the Gaussian integral or the relative logarithm.

In the matrix construction of [2, Theorem F and Corollary 12.1], $D = D_W(s)$ is the covariance on the selected outer coordinates W, with $\|E\| < 1/4$ relative to its common real comparison. The real coordinate count is the outer count on the wall and held guards. The inner fine lattice and the full torus add no dimension to this bound. The small sector budget there, chosen after the localization scale, can in particular pay for the factor M^4 in the relative log determinant. The field and current derivatives above are taken on contained discs in the weighted domains of that construction. For independent cube parameters of radius $\mathcal{S}$, the Cauchy radii may be chosen from the distance to its boundary.

At complete parameters, if A_Y is the precision on the conditioned coordinate domain Y, then $D_W = \chi_W A_Y^{-1} \chi_W^{\mathsf{T}}$. This is a marginal after conditioning on Y. For separated guarded components the constructed covariance is block diagonal. Each block inherits the relative sector: for its coordinate selector J, the map $(JD_0J^{\mathsf{T}})^{-1/2} J D_0^{1/2}$ is a coisometry, and the block relative perturbation is its compression of E. The exponents split, and absolute convergence permits Fubini. The full measure and each block measure have mass one, so their normalizing determinant factors also multiply with the specified branches. Thus the integrals of products of admissible tests factor. These assertions use the prescribed covariance on each declared domain. Independently chosen guarded extensions need not have common marginals away from complete parameters.

5.1. **An auxiliary Gaussian and its local marginal.**

Proposition 5.2 (Eliminating an auxiliary Gaussian)**.** Let $C = C^{\mathsf{T}}$ and Γ be complex matrices of sizes $m \times m$ and $m \times n$. Suppose $C_0 > 0$ is real symmetric, Γ_0 is real, and

$$\|C_0\| + \|C_0^{-1}\| + \|\Gamma_0\| \le K, \qquad \|C - C_0\| + \|\Gamma - \Gamma_0\| \le \varepsilon.$$

Put $D = C + C\Gamma\Gamma^{\mathsf{T}}C$ and define D_0 by the same formula at the real reference. There are positive $\varepsilon_0, \rho_0, C_K$, depending only on K, with the following properties. If $\varepsilon \le \varepsilon_0$ and $|F(u)| \le Ae^{\rho|u|^2}$ for a real-variable Borel test and $0 \le \rho \le \rho_0$, then the joint real-coordinate integral below converges absolutely and satisfies

$$\int d\mu_{I_n}(v)\, e^{-\frac{1}{2}v^{\mathsf{T}}\Gamma^{\mathsf{T}}C\Gamma v} \int e^{u^{\mathsf{T}}\Gamma v} F(u)\, d\mu_C(u) = \int F(u)\, d\mu_D(u). \tag{5.vii}$$

Its value at $F = 1$ is one, and

$$\int |F(u)|\,|d\mu_D(u)| \le Ae^{C_K(\varepsilon+\rho)m}. \tag{5.viii}$$

The dimensions m, n do not enter the constants. ■

Proof. The joint precision on the real variables (u, v) is

$$B = \begin{pmatrix} C^{-1} & -\Gamma \\ -\Gamma^{\mathsf T} & I_n + \Gamma^{\mathsf T} C \Gamma \end{pmatrix}.$$

At the reference its quadratic form is

$$(u - C_0\Gamma_0 v)^{\mathsf T} C_0^{-1}(u - C_0\Gamma_0 v) + |v|^2.$$

The triangular map $(u, v) \mapsto (u - C_0\Gamma_0 v, v)$ and its inverse have norm at most $1 + K^2$. Thus $B_0 \ge c_K I$. The inverse identity for C and expansion of the remaining products give $\|B - B_0\| \le C_K\varepsilon$. Choose the two radii so that this change and 2ρ together are less than $c_K/2$. The real part of the exponent then gives an integrable joint Gaussian majorant. Fubini applies to eq. 5.vii.

Set $A_v = I_n + \Gamma^{\mathsf T} C \Gamma$. It is in a fixed relative precision sector about its real positive reference. Its inverse is in the covariance sector of theorem C after decreasing ε_0. Integrating v first, apply eq. 5.iii with covariance A_v^{-1} and force $\Gamma^{\mathsf T} u$. The result is the factor

$$(\det A_v)^{-1/2} \exp\!\left(\tfrac{1}{2} u^{\mathsf T} \Gamma A_v^{-1} \Gamma^{\mathsf T} u\right).$$

There is no change in the real argument of F. Direct multiplication and the determinant obtained by eliminating either off-diagonal block of $\left(\begin{smallmatrix} I_m & -C\Gamma \\ \Gamma^{\mathsf T} & I_n \end{smallmatrix}\right)$ give

$$D^{-1} = C^{-1} - \Gamma A_v^{-1} \Gamma^{\mathsf T}, \qquad \det D = \det C \, \det A_v.$$

The determinant roots multiply at the real reference. Nonvanishing and connectedness extend that equality to the small neighborhood. The u density after integration is exactly $d\mu_D$. The convention $n = 0$ removes the auxiliary integral and factor.

Since $D_0 \ge C_0$, its inverse norm is at most K; also $\|D_0\| \le C_K$. Expanding $D - D_0$ gives $\|D_0^{-1/2}(D - D_0)D_0^{-1/2}\| \le C_K\varepsilon$. Thus theorem C gives normalization and eq. 5.viii, after decreasing the radii to fit that sector. The bound takes the absolute value after forming the marginal, which accounts for the dimension m. □

Partition u into disjoint coordinate blocks u_b. For a set P of distinct blocks, thresholds $B_b > 0$, and $\gamma \ge 0$ with $\rho + \gamma/2 \le \rho_0$,

$$\int \prod_{b\in P} \mathbf{1}_{\{|u_b|\ge B_b\}} e^{\rho|u|^2} |d\mu_D(u)| \le \exp\left\{ C_K(\varepsilon + \rho + \gamma/2)m - \tfrac{\gamma}{2} \sum_{b\in P} B_b^2 \right\}. \tag{5.ix}$$

On the event in the integrand $|u|^2 \ge \sum_{b\in P} B_b^2$. Insert this inequality and apply eq. 5.viii at growth $\rho + \gamma/2$. The argument needs no independence between the tested coordinates. Fixed Borel restrictions and holomorphic parameter dependence are covered by corollary 5.1. For the rescaled pair $(\tau^{-1}C, \tau^{1/2}\Gamma)$, the effective covariance is $\tau^{-1}D$. Use the positive square root at a real $\tau > 0$ and its nearby continuation. On a fixed compact real interval in $(0, \infty)$ the rescaled reference bounds are uniform, and a sufficiently small complex displacement of τ stays in the same type of matrix neighborhood. When a region has N localization cubes and $m \le aM^4N$, its cost is $\exp\{C_K a(\varepsilon + \rho)M^4N\}$. When the count is already the number m of selected coordinates, that count is used directly in eq. 5.viii. A localization cube of side M in coarse lattice units contains at most $C(LM)^4$ fine bond-colour coordinates. For this fine-coordinate auxiliary marginal one can therefore take $a = CL^4$. This count is distinct from the outer selected-coordinate count in the moving-wall application above.

6. Real Gaussian block restrictions

Let $Z = (Z_b)_{b\in\mathcal{I}}$ be a centered real Gaussian on finitely many Euclidean blocks, with coordinate covariance $\Gamma \geq 0$. It can be represented as $Z = AX$ with X a standard Gaussian in $\mathbb{R}^q$, allowing $q = 0$. For $B_b > 0$, write

$$\mathcal{B} = \prod_b \{|z_b| \leq B_b\}, \qquad \nu = \mathcal{L}(Z \mid |Z_b| < B_b \text{ for every } b).$$

This event has positive probability because its preimage contains a neighborhood of zero. Let $d_b = \operatorname{rank}\Gamma_{bb}$ and $\Gamma_{bb} \leq \sigma_b^2 I$. If $d_b > 0$, take $\sigma_b > 0$.

Lemma 6.1 (Conditional supports and radial weights)**.** For $d_b > 0$, put $V_b = \operatorname{ran}\Gamma_{bb}$. Let A_a be the block row of A with output Z_a. Let W be an independent standard Gaussian on the possibly zero-dimensional space $\ker A_b$, with its induced Euclidean inner product, and set $A'_a = A_a|_{\ker A_b}$. Let Γ_{bb}^+ be the inverse on V_b extended by zero. Conditional on $Z_b = z \in V_b$, the other blocks have a version of their joint law given by

$$Z_a = \Gamma_{ab}\Gamma_{bb}^+ z + A'_a W, \qquad a \neq b, \tag{6.i}$$

For continuous even log-concave functions $0 \leq \psi_a \leq 1$ on the real blocks, the conditional weight

$$w_b(z) = \mathbb{E}_W \prod_{a\neq b} \psi_a(\Gamma_{ab}\Gamma_{bb}^+ z + A'_a W)$$

is even and log-concave on V_b. In particular $w_b(rn) \leq w_b(sn)$ if $r \geq s \geq 0$ and $|n| = 1$ in V_b. ■

Proof. Decompose X orthogonally into the row space of A_b and $\ker A_b$. These projections are independent standard Gaussians on their subspaces. The projection onto the row space is determined by z and equals $A_b^{\mathsf{T}}\Gamma_{bb}^+ z$. Applying A_a gives eq. 6.i. No estimate of the regression matrix is needed.

We use a one-dimensional integration fact. Suppose $f, g, h \geq 0$ are integrable continuous functions on the line, positive on the interiors of their interval supports, and

$$h((1-t)x + ty) \geq f(x)^{1-t} g(y)^t, \qquad 0 < t < 1.$$

If $F = \int f$ and $G = \int g$ are positive, let T be the increasing quantile map from f/F to g/G. In the positive interior its derivative is $T'(x) = f(x)G/(Fg(T(x)))$. Substitute $(1-t)x + tT(x)$ in the integral of h. Its derivative is at least $T'(x)^t$ by the arithmetic-geometric mean inequality. Integrating first on compact intervals in the positive interior and then exhausting it gives

$$\int h \geq \int f(x)^{1-t} g(T(x))^t T'(x)^t \, dx = F^{1-t} G^t.$$

If either integral vanishes, the same conclusion holds directly. Applying this fact to slices of a jointly log-concave function shows that integration in one coordinate preserves log-concavity. The positive sets of these slices are intervals, and the slices and their marginals are continuous by domination by a Gaussian in the integrated variables. Iterating one coordinate at a time therefore proves the marginal claim for the product of the cutoffs and the Gaussian density of W. That product is log-concave by affine substitution and multiplication. It is invariant under $(z, W) \mapsto (-z, -W)$. Hence w_b is even. Writing sn as a convex combination of rn and $-rn$ gives the ray inequality, including when $w_b(rn) = 0$. □

For $0 < \delta \leq 1/2$, use smooth radial cutoffs $\psi_{B,\delta}$ that equal one on $|z| \leq (1-\delta)B$ and vanish on $|z| \geq B$. To construct them, put $\phi = 0$ on $(-\infty, 0]$, take

$$\phi(t) = \int_0^t (t-s) e^{-1/s^2} (1-s)^{-4} \, ds \quad (0 < t < 1), \qquad \phi = +\infty \text{ on } [1, \infty),$$

and set $\psi_{B,\delta}(z) = \exp[-\phi((|z|/B - 1 + \delta)/\delta)]$. The function ϕ is nondecreasing and convex. It is flat at zero and grows at least as $c(1-t)^{-2}$ at one. Its derivatives at one grow at most as inverse

powers of $1-t$, so every derivative of the exponential vanishes there. This proves smoothness of the cutoff at its two transition spheres. Convex composition with the norm gives log-concavity. As $\delta \downarrow 0$, the argument of ϕ decreases at every fixed $|z| < B$, so the cutoffs increase to the strict-ball indicator. Put

$$\chi_\delta = \prod_b \psi_{B_b,\delta}(Z_b), \qquad \nu_\delta[P] = \frac{\mathbb{E}[P(Z)\chi_\delta]}{\mathbb{E}\chi_\delta}.$$

The denominators are positive. For each polynomial, dominated convergence gives $\nu_\delta[P] \to \nu[P]$.

Proposition 6.2 (Normalized cutoff flux)**.** For each block define

$$\beta_{b,\delta} = \frac{\mathbb{E}[|\nabla\psi_{B_b,\delta}(Z_b)| \prod_{a\neq b} \psi_{B_a,\delta}(Z_a)]}{\mathbb{E}\chi_\delta}. \tag{6.ii}$$

For $d_b = 0$ this is zero. If $d_b > 0$, then

$$\beta_{b,\delta} \le \beta_b^* = \frac{C_{d_b}}{B_b}\left[1 + (B_b/\sigma_b)^{d_b}\right] e^{-c_{d_b}B_b^2/\sigma_b^2}. \tag{6.iii}$$

Take $\beta_b^* = 0$ at rank zero. These bounds are independent of the other blocks, of their number and dependencies, and of any lower marginal eigenvalue. If $B_b \ge \sigma_b > 0$ the right side is at most $2C_{d_b}B_b^{d_b-1}\sigma_b^{-d_b}e^{-c_{d_b}B_b^2/\sigma_b^2}$. ■

Proof. A rank-zero block is identically zero and its cutoff is constant there. For positive rank use lemma 6.1 and polar coordinates $z = rn$ in V_b. Its intrinsic Gaussian density in direction n, omitting a common determinant constant, is $r^{d_b-1}e^{-q(n)r^2/2}$, where

$$q(n) = n^{\mathsf{T}}(\Gamma_{bb}|_{V_b})^{-1}n \ge \sigma_b^{-2}.$$

The radial gradient of the cutoff is also intrinsic in V_b. Set $B = B_b$, $a = (1-\delta)B$, and $d = d_b$. Let D_n denote the integral in r of this density times $\psi_{B,\delta}(rn)w_b(rn)$. On the plateau the cutoff is one. For every $r \in [a, B]$ ray monotonicity gives

$$D_n \ge w_b(rn)\, I(q(n), a), \qquad I(q,a) = \int_0^a s^{d-1}e^{-qs^2/2}\, ds.$$

The negative radial derivative of the cutoff has integral one on $[a, B]$. Thus the flux numerator in this direction is at most $D_n B^{d-1}e^{-q(n)a^2/2}/I(q(n), a)$. This also holds if $D_n = 0$.

Integrating to $\min(a, q^{-1/2})$ shows $I(q,a) \ge e^{-1/2}\min(a, q^{-1/2})^d/d$. Since $a \ge B/2$, the ratio is at most $C_d B^{-1}[1+(qB^2)^{d/2}]e^{-qB^2/8}$. For $qB^2 \ge B^2/\sigma_b^2$, absorb the polynomial at qB^2 into half the exponential. This gives the bound in eq. 6.iii. Integration in n cancels the common determinant constant and the integral of D_n, which proves the normalized estimate. □

The B_b^{-1} term matters at small radius. For an isotropic d-dimensional block, the sharp radial boundary density divided by the ball probability is asymptotic to d/B_b as $B_b/\sigma_b \to 0$. This follows by substituting $r = B_b t$ in the two radial integrals.

Corollary 6.3 (Integration by parts at a sharp block restriction)**.** For a linear statistic $L = \sum_i \ell_i^{\mathsf{T}} Z_i$ and polynomial P, let

$$\mathcal{D}_L(P) = \nu[LP] - \sum_{i,b} \ell_i^{\mathsf{T}} \Gamma_{ib} \nu[\nabla_b P]. \tag{6.iv}$$

Then

$$|\mathcal{D}_L(P)| \le \|P\|_{\infty,\mathcal{B}} \sum_{i,b} |\ell_i| \|\Gamma_{ib}\| \beta_b^*.$$

For an empty block set all sums and the nonnegative budgets below are zero. In particular, with positive amplitudes a_b and cell measures m_b, set

$$S = \sup_i \sum_b m_b \frac{\|\Gamma_{ib}\|}{a_i a_b}, \qquad \varepsilon_\partial = \sup_b \frac{a_b \beta_b^*}{m_b}.$$

The bound is at most $S\varepsilon_\partial \|P\|_{\infty,\mathcal{B}} \sum_i a_i |\ell_i|$. ■□

Proof. Apply Gaussian integration by parts in X to $P(AX)$ times the smooth product cutoff. The two factors of A give $AA^{\mathsf{T}} = \Gamma$, regardless of its rank. The cutoff derivative at block b contributes $\ell_i^{\mathsf{T}} \Gamma_{ib}$ times

$$\frac{\mathbb{E}[P(Z) \nabla \psi_{B_b,\delta}(Z_b) \prod_{a \ne b} \psi_{B_a,\delta}(Z_a)]}{\mathbb{E}\chi_\delta}.$$

Its norm is bounded by $\|P\|_{\infty,\mathcal{B}} \beta_b^*$, since the derivative and other cutoffs restrict to $\mathcal{B}$. The polynomial terms converge to the two terms of eq. 6.iv. Thus the total defect has the claimed bound, even if separate boundary contributions of coincident constraints have no intrinsic sharp limit. Insert $a_i m_b/(a_i a_b)$ and the compensating factor a_b/m_b in the displayed sum to obtain the weighted bound. □

Here Γ_{ib} is a coordinate covariance. If a positive definite precision K is self-adjoint for the weighted pairing with matrix $M = \operatorname{diag}(m_b I)$, the coordinate density uses precision MK. When the inverse has physical kernel G, the array identity is $K^{-1} = \mathsf{G}M$, so $(MK)^{-1} = \mathsf{G}$ is the coordinate covariance. This explains the factor m_b^{-1} in the boundary budget.

Corollary 6.4 (Linear statistics under block restrictions)**.** For a real linear statistic L with unconditioned variance v^2, and any nonnegative increasing function f on $[0,\infty)$,

$$\nu[f(|L|)] \le \mathbb{E} f(v|N|), \qquad N \sim N(0,1).$$

In particular $\nu[L^{2m}] \le (2m-1)!! v^{2m}$, $\nu[e^{tL}] \le e^{t^2 v^2/2}$ for real t, and $\operatorname{Cov}_\nu(Z) \le \Gamma$. ■□

Proof. If $v = 0$, both sides use $L = 0$. Otherwise condition X on the single linear coordinate $L = x$, using the orthogonal decomposition in lemma 6.1. With all smooth cutoffs included, the resulting weight on the one-dimensional Gaussian density is even and nonincreasing in $|x|$. Let R, R' be independent copies of $v|N|$. The product $(f(R) - f(R'))(w(R) - w(R'))$ is nonpositive. For bounded f, take expectations and divide by $\mathbb{E}w(R) > 0$ to get the claimed inequality for ν_δ. Pass to ν by dominated convergence and then to unbounded f by truncation. The entire restricted law is even. Apply the inequality to powers and to $\cosh(t|L|)$, and use the Gaussian moment identities. Testing all real linear statistics proves the covariance order. □

7. A holomorphic constrained coordinate chart

A complex constraint does not define a real delta measure by substitution into δ_F. We instead continue a specified real fibre chart. The normalized Gaussian theorems then act on its fixed real coordinates. The chart Jacobian remains part of the exact, unnormalized block integral.

Let $Q_0 : E \to F$ be an onto real map, let $K_0 = \ker Q_0$, and let $L_0 = Q_0^*(Q_0 Q_0^*)^{-1}$. Write $i_0 : K_0 \to E$ for inclusion. Suppose $Q(z), T(z)$ depend holomorphically on a parameter domain

containing a real reference z_0, with $Q(z_0) = Q_0$ and $T(z_0)$ real symmetric and positive on K_0. Transposes below are bilinear continuations. Assume $A(z) = Q(z)L_0$ is invertible and define

$$L(z) = L_0 A(z)^{-1}, \qquad Z(z) = (I - L(z)Q(z))i_0, \qquad \mathcal{T}(z) = Z(z)^{\mathsf{T}} T(z) Z(z). \tag{7.i}$$

Then $QL = I$ and $QZ = 0$. Relative to the decomposition $E = i_0 K_0 \oplus L_0 F$, the map $(u, b) \mapsto Zu + Lb$ has triangular blocks $\left(\begin{smallmatrix} I & 0 \\ -A^{-1}Qi_0 & A^{-1} \end{smallmatrix}\right)$. Its continued coarea density is therefore

$$j(z) = J_{Q_0}^{-1} \det A(z)^{-1}. \tag{7.ii}$$

The orientation is chosen so that $j(z_0) > 0$. Where $Q(z)$ is real and $\det A(z) > 0$, this is the real measure factor in the u coordinate. It is not obtained by putting an absolute value into a complex determinant.

Proposition 7.1 (A constrained integral in fixed coordinates)**.** In eq. 7.i, suppose $T(z)^{\mathsf{T}} = T(z)$ and that $\mathcal{T}(z)^{-1}$ lies in the relative covariance sector of theorem C about $\mathcal{T}(z_0)^{-1}$. For $b \in F_{\mathbb{C}}$ define

$$\mathcal{I}(z, b) = j(z) \int_{K_0} \exp\{-\tfrac{1}{2}(Zu + Lb)^{\mathsf{T}} T(z)(Zu + Lb)\}\, du,$$

where u is real. Put

$$\begin{aligned} H(z) &= L - Z\mathcal{T}^{-1}Z^{\mathsf{T}}TL, & C(z) &= Z\mathcal{T}^{-1}Z^{\mathsf{T}}, \\ \Sigma(z) &= L^{\mathsf{T}}TL - L^{\mathsf{T}}TZ\mathcal{T}^{-1}Z^{\mathsf{T}}TL. \end{aligned} \tag{7.iii}$$

Then

$$\mathcal{I}(z, b) = j(z)(2\pi)^{\dim K_0/2} \det(\mathcal{T}(z))^{-1/2} e^{-b^{\mathsf{T}}\Sigma(z)b/2}, \tag{7.iv}$$

with the positive-reference branch. The integral is holomorphic in z and entire in b. Its normalized exponential test is

$$\frac{j(z)}{\mathcal{I}(z, b)} \int_{K_0} e^{\ell^{\mathsf{T}}(Zu+Lb)} e^{-(Zu+Lb)^{\mathsf{T}}T(Zu+Lb)/2}\, du = e^{\ell^{\mathsf{T}}Hb + \ell^{\mathsf{T}}C\ell/2}. \tag{7.v}$$

It gives all polynomial tests by differentiation in $\ell \in E_{\mathbb{C}}$. In particular $QH = I$, $QC = 0$, and $Z^{\mathsf{T}}TH = 0$. ■

Proof. The inverse covariance sector gives a strictly positive real part of $u^{\mathsf{T}}\mathcal{T}u$, uniformly on compact parameter sets. On compact sets of b the cross term is bounded by a constant times $|u|$. Gaussian domination therefore proves absolute convergence and holomorphic parameter dependence. Write $g = Z^{\mathsf{T}}TLb$. The exponent is $-u^{\mathsf{T}}\mathcal{T}u/2 - g^{\mathsf{T}}u - b^{\mathsf{T}}L^{\mathsf{T}}TLb/2$. Apply eq. 5.iii to the fixed real variable u with covariance $\mathcal{T}^{-1}$ and linear force $-g$. This proves eq. 7.iv. With the additional test, the force becomes $Z^{\mathsf{T}}\ell - g$ and the constant term gains $\ell^{\mathsf{T}}Lb$. Expanding this quadratic expression proves eq. 7.v. No integration variable was shifted by a complex vector. The three matrix identities follow directly from $QL = I$, $QZ = 0$ and the definition of H. □

When Q, T, b are real and $\det A > 0$, this chart integral is the original constrained integral. The untested formula is independent of the chosen splitting on the common connected parameter component containing the reference, when the branches agree at the reference. Indeed another kernel basis and right inverse have the form $Z' = ZM$ and $L' = L + ZX$, with M invertible and real at the reference. Put $\epsilon_M = \operatorname{sgn}\det M(z_0)$. The positive-reference coordinate density is multiplied by $\epsilon_M \det M$, while the restricted quadratic form becomes $M^{\mathsf{T}}\mathcal{T}M$. Its positive-reference continued square root is multiplied by the same $\epsilon_M \det M$. The Schur form Σ is unchanged by the shear X. These algebraic identities prove agreement on that component. On any other component of the overlap, the transition branches must be matched separately; agreement at the reference

alone does not determine their relative sign there. No open set of real points on a parameter curve is needed. The exact determinant identity, including the chart density, is

$$j(z)^{-2}\det \mathcal{T}(z) = (-1)^{\dim F}\det \mathbb{S}(T(z), Q(z)).$$

It follows by the same saddle congruence as eq. 2.ii, using $QZ = 0$ and $QL = I$. The right side is positive at the real reference. The sign of a real saddle determinant must be accounted for before selecting its positive-reference square root.

A fixed Borel test may be integrated on the real u coordinates. At nonzero complex datum the normalized density in this chart is the tilted measure

$$e^{-g^{\mathsf{T}}u - g^{\mathsf{T}}\mathcal{T}^{-1}g/2} d\mu_{\mathcal{T}^{-1}}(u), \qquad g = Z^{\mathsf{T}}TLb. \tag{7.vi}$$

The centered estimate cannot be used without this tilt. For a test bounded by $F_0 e^{\rho|u|^2}$ with $F_0 \geq 0$ and $0 \leq \rho \leq (8K)^{-1}$, put $D = \mathcal{T}^{-1}$ and $A_\rho = \operatorname{Re} D^{-1} - 2\rho I$. The sector proof gives $A_\rho > 0$ and $\|A_\rho^{-1}\| \leq 12K/5$. After taking the absolute value, completion of a real square gives the additional factor

$$\exp\Big\{-\tfrac{1}{2}\operatorname{Re}(g^{\mathsf{T}}Dg) + \tfrac{1}{2}(\operatorname{Re} g)^{\mathsf{T}}A_\rho^{-1}(\operatorname{Re} g)\Big\}.$$

It is at most $e^{CK|g|^2}$ because $\|D\| \leq 5K/4$. Thus the normalized absolute integral is bounded by $F_0 \exp\{CK|g|^2 + C(r + K\rho)\dim K_0\}$, including the endpoint $\rho = (8K)^{-1}$. Uniform bounds on a parameter polydisc give the Cauchy derivative estimates in the same way. A Gaussian defined instead on the centered chart $x = Hb + Zv$ has the centered measure in its declared real v coordinate. Arbitrary Borel tests on these two complex charts are not identified. They cannot be evaluated at complex-shifted points without an extension and a proof of the resulting formula. The finite determinant $j(z)$ is retained in the unnormalized integral. A cost per localization cube for a normalized selected Gaussian is the separate dimension bound in corollary 5.1; the chart identity alone gives no locality estimate for a global coarea determinant.

7.1. **The physical normalizer in complex parameters.**

Corollary 7.2 (Complex physical parameters)**.** Suppose the data of proposition 4.2 extend holomorphically to a connected physical parameter chart containing an open neighborhood of V_* in its underlying real coordinate slice. Require its real hypotheses on that neighborhood. Assume the fibre Gaussians have the fixed-coordinate representations and relative covariance sectors of proposition 7.1, with their transition branches matched, and that the signed saddle determinants below are nonzero and have holomorphic logarithms continued from their positive real reference values. With bilinear physical transposes, put

$$s_{\mathrm{f}}(V) = (-1)^{\dim F}\det \mathbb{S}(B_{\mathrm{fib}} + D(I - R^{\mathsf{T}}R)D^{\mathsf{T}}, Q),$$
$$s_{\mathrm{s}}(V) = (-1)^{\dim \mathcal{K}}\det \mathbb{S}(D^{\mathsf{T}}D, R).$$

Then the normalized physical Gaussian logarithm is

$$\Psi(V) = -\frac{1}{2}\log\frac{s_{\mathrm{f}}(V)}{s_{\mathrm{f}}(V_*)} + \log\frac{s_{\mathrm{s}}(V)}{s_{\mathrm{s}}(V_*)},$$

with both logarithms zero at the reference. The parameters are those of the physical chart; the conclusion asserts no extension to independent currents off that chart. ■□

Proof. The fixed-coordinate Gaussian formula and its domination argument in proposition 7.1 prove holomorphy of the normalized integral; that argument uses only the stated covariance sector. Exponentiating the displayed signed saddle expression gives a holomorphic function on the chart. On a small connected real neighborhood of V_*, $\det A$ is real, nonzero and has a fixed sign. The continued chart integral differs from the positive real integral by this fixed sign, which cancels on normalization at V_*. Proposition 4.2 therefore identifies the normalized chart integral with the exponential of

the displayed saddle expression. An equality of holomorphic functions on an open real coordinate neighborhood extends to a complex neighborhood: all real partial derivatives of the difference vanish there, and hence so do its complex Taylor coefficients. The identity theorem on the connected chart proves their equality. The displayed expression therefore supplies a holomorphic logarithm of this normalized integral, equal to zero at the reference. Nonvanishing of the integral alone would not have guaranteed such a global logarithm. □

8. The free action-density covariance

Use the Hermitian colour convention of Section 2 and write $d_G = 3$ for its real dimension. The calculation below uses d_G independent scalar colour components. Let $g > 0$ be the coupling for which g^{-2} multiplies the Wilson action in the exponent. On a raw link field A, the coboundary $\mathrm{d}A$ is its unscaled oriented plaquette sum. The quadratic action at the identity is

$$\frac{1}{2g^2}\|\mathrm{d}A\|_2^2$$

in counting coordinates. Indeed the logarithm of the plaquette product of e^{itA_b} has first term $it(\mathrm{d}A)(p)$ and has trace zero. Thus the t^2 coefficient of $1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U_p(t)$ is $\frac{1}{2}|(\mathrm{d}A)(p)|^2$. This agrees with the flat curl form of [5, eqs. (3.10)–(3.12), p. 392].

8.1. **The continuum curvature and its contractions.** Use the Fourier transform $\widehat{f}(k) = \int e^{-ik\cdot x} f(x)\,dx$ with inverse factor $(2\pi)^{-4}$. Define the potential as four independent centred Gaussian test families per colour, with covariance $g^2 G$ in each family, where

$$\widehat{G}(k) = |k|^{-2}, \qquad G(x) = \int_0^\infty (4\pi t)^{-2} e^{-|x|^2/(4t)}\,dt = \frac{1}{4\pi^2|x|^2} \quad (x \neq 0).$$

The heat integral follows by Fourier inversion of the Gaussian and the identity $\int_0^\infty e^{-t|k|^2}\,dt = |k|^{-2}$. Both identities hold against Schwartz tests by absolute integration, using their decay and the local integrability of $|x|^{-2}$ and $|k|^{-2}$ in four dimensions. The covariance form is positive because it is the L^2 pairing of the Fourier tests multiplied by $|k|^{-1}$. In the separable real Hilbert completion for this form, summing independent standard normals against the orthonormal coordinates of a test constructs the family in L^2.

The curvature is $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ in the sense of test families. Write K for its unit-coupling covariance distribution in one colour and use the orthonormal basis $e_\mu \wedge e_\nu$, $\mu < \nu$, on $\Lambda^2\mathbb{R}^4$. For $P = (\mu,\nu)$ and $Q = (\rho,\sigma)$ the distribution is

$$\begin{aligned}\mathsf{K}_{PQ} = {} & -\delta_{\nu\sigma}\partial_\mu\partial_\rho G + \delta_{\nu\rho}\partial_\mu\partial_\sigma G \\ & + \delta_{\mu\sigma}\partial_\nu\partial_\rho G - \delta_{\mu\rho}\partial_\nu\partial_\sigma G.\end{aligned} \tag{8.i}$$

One derivative acts in each covariance argument, which gives the minus sign on the first term.

Lemma 8.1 (The curvature tensor away from coincidence)**.** For $u \neq 0$, put $\widehat{u} = u/|u|$ and $R_u = I - 2\widehat{u}\widehat{u}^{\mathsf{T}}$. Then

$$\mathsf{K}(u) = \pi^{-2}|u|^{-4}\Lambda^2 R_u, \qquad \|\mathsf{K}(u)\|_{\mathrm{HS}}^2 = 6\pi^{-4}|u|^{-8}.$$

Here $\|\cdot\|_{\mathrm{HS}}$ is the Hilbert–Schmidt norm in the six-dimensional two-form basis. ■

Proof. The Hessian of G is $(4\widehat{u}\widehat{u}^{\mathsf{T}} - I)/(2\pi^2|u|^4)$. For a symmetric matrix M, its action on each of the two factors of a wedge, added together, sends $v \wedge w$ to $Mv \wedge w + v \wedge Mw$. Equation 8.i is the negative of this action for $M = \nabla^2 G$. Since $\widehat{u}\widehat{u}^{\mathsf{T}}$ has rank one, the product term with this matrix acting on both factors vanishes. Expanding $\Lambda^2 R_u$ therefore gives the stated identity. The reflection is orthogonal, as is its two-form action. □

To define the separated density correlations, fix an even, nonnegative smooth function χ of integral one supported in the unit ball and set $\chi_\varepsilon(x) = \varepsilon^{-4}\chi(x/\varepsilon)$. Let $F_\varepsilon(x)$ be the curvature tested against its translate to x. Use the centred density

$$q_\varepsilon(x) = \frac{1}{2}\sum_{\alpha=1}^{d_G}\sum_{P}\bigl(F^{\alpha}_{\varepsilon,P}(x)^2 - \mathbb{E}[F^{\alpha}_{\varepsilon,P}(x)^2]\bigr).$$

For each $\varepsilon > 0$, integrate this field against continuous compactly supported tests by L^p Riemann sums, $1 \le p < \infty$. The smoothed covariance is continuous. Gaussian moment bounds then give L^{2p} continuity of each curvature component and L^p continuity of its square.

Proposition 8.2 (Separated connected Gaussian densities)**.** For $n \ge 2$, the mixed cumulant of the regularized density has a limit on every collection of pairwise separated compact supports. For noncoincident points $x_1, \dots, x_n$ its kernel is

$$C_n(x_1, \dots, x_n) = \frac{d_G g^{2n}}{2n}\sum_{\sigma \in S_n}\operatorname{tr}\left(\prod_{j=1}^{n}\mathsf{K}(x_{\sigma(j)} - x_{\sigma(j+1)})\right), \qquad \sigma(n+1) = \sigma(1). \tag{8.ii}$$

The product follows the displayed order. In particular the two-point smearing limit is $C_G g^4 \mathcal{K}(f,h)$, where

$$C_G = \frac{3d_G}{\pi^4} = \frac{9}{\pi^4}, \qquad \mathcal{K}(f,h) = \iint f(x)h(y)|x-y|^{-8}\,dx\,dy. \tag{8.iii}$$

Every odd kernel C_n, $n \ge 3$, vanishes on the separated configuration space. At four distinct collinear points,

$$C_4(x_1,x_2,x_3,x_4) = \frac{6d_G g^8}{\pi^8}\sum_{\text{three unoriented cycles } \{i,j\} \text{ in the cycle}}\ \prod |x_i - x_j|^{-4}. \tag{8.iv}$$

This is strictly positive and remains positive on the product of sufficiently small disjoint neighborhoods of the four points. ∎

Proof. For any real finite Gaussian vector with covariance C, write it as $C^{1/2}$ times a standard Gaussian. If T_i are real symmetric matrices, the logarithmic moment generating function of its centred half-quadratic forms is

$$-\frac{1}{2}\log\det(I - C\textstyle\sum_i z_i T_i) - \frac{1}{2}\sum_i z_i \operatorname{tr}(CT_i).$$

Diagonalizing the symmetric quadratic form for small real z and integrating each coordinate proves the formula. Sylvester's determinant identity also covers singular C. The resulting power series at zero is obtained by integrating the convergent geometric series for the inverse in the derivative of the logarithm. For $n \ge 2$, the mixed derivative of order n comes from its term $\operatorname{tr}((C\sum_i z_i T_i)^n)/(2n)$. For curvature at n points use the coordinate projections for the T_i. Each permutation gives one ordered cycle of covariance blocks. Independence multiplies the expression by d_G.

For separated supports the smoothed cross-covariance blocks converge uniformly, with every fixed derivative, to $g^2\mathsf{K}$. Indeed their convolutions sample a fixed compact neighborhood away from coincidence, on which each derivative of the kernel is uniformly continuous. Differentiation under the convolution and the unit integral of the mollifier give the asserted convergence. Integrating the finite cycle formula and taking this limit proves eq. 8.ii. For $n = 2$ the two permutations give the squared Hilbert–Schmidt norm with prefactor $d_G g^4/2$. Lemma 8.1 gives eq. 8.iii.

On two-forms the Hodge star J satisfies $J^2 = I$ and $J\Lambda^2 R = (\det R)\Lambda^2 R\, J$ for R orthogonal. This follows by applying R to the defining wedge identity for the star and tracking its action on the oriented volume. Each $\mathsf{K}(u)$ thus anticommutes with J, so conjugation by J negates an odd product and preserves its trace. At collinear points all the reflections agree, since $R_u = R_{-u}$. The product of four two-form actions is the identity, whose trace is six. Each unoriented four-cycle

has eight representatives in S_4, giving eq. 8.iv. Smoothness at separated arguments then gives positivity on the stated neighborhoods. □

The separated limits concern mixed cumulants. In fact for a nonnegative nonzero test f, the variance of $\int f(x)q_\varepsilon(x)\,dx$ grows at least as $c_f g^4\varepsilon^{-4}$. Choose a ball on which f has a positive lower bound. In its interior, and for $M\varepsilon \le |x-y|$ with M sufficiently large, the convolution error in $\mathsf{K}(x-y)$ is at most $C\varepsilon|x-y|^{-5}$, by its first derivative. Its squared Hilbert–Schmidt norm is thus bounded below by $c|x-y|^{-8}$. The Wick variance integrates this bound on a shell with fixed outer radius inside the ball. Its radial integral is a positive multiple of $\int_{M\varepsilon}^{r_0} r^{-5}\,dr$.

8.2. **The lattice projection and its physical kernel.** Let $a>0$, $\ell = ma$ with m a positive integer, and $\Lambda_{a,\ell} = a\mathbb{Z}^4/\ell\mathbb{Z}^4$. The coordinates are based positive bonds and plaquettes. Write $p=(x,P)$, $P=(\mu,\nu)$ with $\mu<\nu$, and $x_p = x + ac_P$, where $c_P = (e_\mu+e_\nu)/2$. Use the exact-form projection P_{ex} of Section 3.1 in one colour, also on lattices with no blocking step. On $(\ker \mathrm{d})^\perp$ the map d is a linear isomorphism onto $\mathcal{E}$. It sends the normalized Gaussian for the raw quadratic action to the Gaussian with covariance g^2P_{ex}. The change of coordinates has a constant Jacobian, which cancels in this normalized law.

For independent raw colour curvatures F_p^α with this covariance, their physical values are $a^{-2}F_p^\alpha$. The physical unit-coupling covariance entry is consequently

$$\mathsf{K}_{a,\ell}(p,q) = a^{-4}P_{\mathrm{ex}}(p,q). \tag{8.v}$$

The centred quadratic observable for a real test f on the torus is

$$Q_a(f) = \frac{1}{2}\sum_{\alpha=1}^{d_G}\sum_p f(x_p)((F_p^\alpha)^2 - \mathbb{E}[(F_p^\alpha)^2]). \tag{8.vi}$$

In physical coordinates the a^4 integration weight cancels the a^{-4} in the square of the physical curvature. The finite Gaussian calculation in Proposition 8.2, with the T_i given by diagonal test multipliers on plaquettes, yields

$$\mathrm{Cov}(Q_a(f),Q_a(h)) = \frac{d_G g^4}{2}\,a^8\sum_{p,q} f(x_p)h(x_q)\mathsf{K}_{a,\ell}(p,q)^2. \tag{8.vii}$$

For a mixed cumulant of order n, the same calculation gives a sum over the permutations in eq. 8.ii and all $p_1,\dots,p_n$. Its summand has $a^{4n}\prod_i f_i(x_{p_i})$ times the product of the entries $\mathsf{K}_{a,\ell}(p_{\sigma(j)},p_{\sigma(j+1)})$. The prefactor is $d_G g^{2n}/(2n)$. These are real scalar entries, including when the plane labels differ.

Let $\delta = 2\pi/\ell$ and choose the finite frequency set $\mathcal{B}_a = \delta\mathbb{Z}^4 \cap [-\pi/a,\pi/a)^4$. For $k\in\mathcal{B}_a$ put

$$t_{a,\mu}(k) = a^{-1}(e^{iak_\mu}-1), \qquad \omega_a(k) = |t_a(k)|^2, \qquad B_a(k)v = t_a(k)\wedge v.$$

The symbol B_a represents the difference $a^{-1}\mathrm{d}$, which has the same range as d. The wedge norm identity gives $B_a^\dagger B_a = \omega_a I - t_a t_a^\dagger$. Here $\dagger$ is the Hermitian adjoint of a Fourier matrix. As $B_a t_a = 0$, the projection symbol is

$$\Pi_a(k) = B_a(k)B_a(k)^\dagger/\omega_a(k) \quad (k\neq 0), \qquad \Pi_a(0) = 0. \tag{8.viii}$$

The rank is three at each nonzero frequency. A constant two-form has zero exact component, including on the one-site torus. The unitary discrete Fourier transform has factor m^{-2} in both directions. Inserting it in eq. 8.v gives, for $p=(x,P)$ and $q=(y,Q)$,

$$\mathsf{K}_{a,\ell}(p,q) = \left[\ell^{-4}\sum_{k\in\mathcal{B}_a} e^{ik\cdot(x-y)}\Pi_a(k)\right]_{PQ}. \tag{8.ix}$$

In particular the finite Fourier coefficient is $a^{-4}m^{-4} = \ell^{-4}$.

The continuum multiplier $\Pi_{\mathrm{cont}}(k)$ is eq. 8.viii with $t_0 = ik$ and $\omega_0 = |k|^2$, and $\Pi_{\mathrm{cont}}(0)=0$ on the reciprocal grid. Its inverse transform on $\mathbb{R}^4$ is eq. 8.i. On the torus write K_ℓ for the distribution

with Fourier series $\ell^{-4}\sum_{k\in\delta\mathbb{Z}^4} e^{ik\cdot z}\Pi_{\mathrm{cont}}(k)$. The zero-frequency convention matters even away from coincidence. The traces of the raw lattice matrix in eq. 8.ix and of K_ℓ are, respectively,

$$3(a^{-4}\mathbf{1}_{\{z=0\}}-\ell^{-4}), \qquad 3(\delta_{\mathrm{per}}(z)-\ell^{-4}).$$

These identities follow by taking the trace of the projection symbol. Here $z=x-y$ and δ_{per} is the unit point mass at the origin of the torus.

Proposition 8.3 (Separated comparison of curvature kernels)**.** Let r be the Euclidean distance on the torus from x_p to x_q. For $r\ge 10a$, if u is the short displacement from x_q to x_p, then

$$|\mathsf{K}_{a,\ell}(p,q)-\mathsf{K}_{\ell,PQ}(u)|\le Cr^{-4}(a/r)^{1/2}. \tag{8.x}$$

For $10a\le r\le \ell/10$ one has

$$|\mathsf{K}_{a,\ell}(p,q)-\mathsf{K}_{PQ}(u)|\le Cr^{-4}\big[(a/r)^{1/2}+(r/\ell)^4\big]. \tag{8.xi}$$

The constant is independent of the orientation labels, the mesh and the period. The smooth torus kernel in eq. 8.x is periodic in u. ∎

Proof. We first obtain the continuum torus estimates. Its heat kernel is the sum of translates by $\ell\mathbb{Z}^4$ of $p_t(x)=(4\pi t)^{-2}e^{-|x|^2/(4t)}$, since integrating this sum against each Fourier mode gives $\ell^{-4}e^{-t|k|^2}$. Integrating the heat kernel with its mean ℓ^{-4} subtracted gives the periodic Green distribution G_ℓ of zero mean. Inserting G_ℓ into eq. 8.i produces K_ℓ.

At period one and $|z|\le 1/4$, the difference G_1-G is smooth. On $0<t\le 1$ all image terms other than the central one, with any fixed number of derivatives, have an integrable bound $Ct^{-M}e^{-c/t}$ for suitable M. This follows from the lower bound $|z+m|\ge c|m|$ on each noncentral image. On $t\ge 1$, the nonzero torus Fourier modes decay exponentially with all fixed derivatives, and the corresponding free heat derivatives are bounded by $C_\alpha t^{-2-|\alpha|/2}$. The mean subtraction over $(0,1]$ is constant. Thus the difference has bounded derivatives of all fixed orders on the ball. On the rest of the unit torus away from zero the same split gives smoothness and bounded derivatives on closed sets. Scaling $G_\ell(z)=\ell^{-2}G_1(z/\ell)$ shows, with $r_z=\mathrm{dist}(z,\ell\mathbb{Z}^4)>0$, that

$$\begin{aligned}\|\partial^\alpha \mathsf{K}_\ell(z)\|&\le C_\alpha r_z^{-4-|\alpha|}\quad (r_z>0),\\ \|\partial^\alpha(\mathsf{K}_\ell-\mathsf{K})(z)\|&\le C_\alpha \ell^{-4-|\alpha|}\quad (0<|z|\le \ell/4).\end{aligned} \tag{8.xii}$$

We compare the Fourier sums at the raw displacement $z=x-y$. The bound $|t_a(k)|\asymp|k|$ on the Brillouin cube follows from the sine formula for $|t_a|$. Differentiating t_a gives $|\partial^\alpha t_a|\le C_\alpha|k|^{1-|\alpha|}$ away from zero, since $a|k|$ is bounded on that cube. The product and quotient rules in eq. 8.viii therefore give $\|\partial^\alpha\Pi_a(k)\|\le C_\alpha|k|^{-|\alpha|}$. The same estimates hold for Π_{cont}. Taylor expansion of t_a yields $\|\Pi_a(k)-\Pi_{\mathrm{cont}}(k)\|\le Ca|k|$ on $a|k|\le c_0$, for a fixed small numerical c_0.

Choose a smooth frequency cutoff equal to one for $|k|\le R$ and zero for $|k|\ge 2R$. Assume for now that $R\ge 18\delta$ and $2aR\le c_0$. On its support the two reciprocal grids coincide. Counting disjoint cubes of side δ around the grid points shows that the low-frequency difference has norm at most CaR^5, with the Fourier factor ℓ^{-4}. For the high-frequency lattice symbol, apply the ninth forward frequency difference in a coordinate μ with $|z_\mu|\ge r_z/2$ in a short representative. The cutoff extends periodically on the Brillouin torus. The derivative estimates above and the product rule bound this difference at k by

$$C\delta^9\big(R+\mathrm{dist}(k,(2\pi/a)\mathbb{Z}^4)\big)^{-9}.$$

Indeed the difference is the integral of the ninth derivative over nine shifts of length at most δ, and R dominates the sum of those lengths. Summation by parts has factor $(e^{-i\delta z_\mu}-1)^9$ on the Fourier sum. The raw z/a is integral, so the sum has periodic phase on this frequency torus. The modulus of the factor is at least $(2r_z/\ell)^9$, by the sine bound for $|z_\mu|\le\ell/2$. The ninth difference has absolute sum at most $C\delta^9R^{-5}$ after multiplying by ℓ^{-4}. This follows by comparing the grid sum of $(R+|k|)^{-9}$ with its integral. The same argument applies to the high-frequency continuum symbol

on the full reciprocal grid. For its distributional Fourier series the multiplication by $(e^{-i\delta z_\mu}-1)^9$ gives an absolutely convergent series of ninth differences. Dividing away from the zero set of this factor proves the pointwise bound there. Both high-frequency terms are thus at most $Cr_z^{-9}R^{-5}$.

Take $R = Dr_z^{-1}(r_z/a)^{1/10}$ with a fixed D large enough that $R \geq 18\delta$, since $r_z \leq \ell$. For r_z/a above a fixed numerical threshold the support condition $2aR \leq c_0$ holds. The sum of the two errors is bounded by $Cr_z^{-4}(a/r_z)^{1/2}$. For the remaining nonzero raw displacements, the lattice Fourier sum has norm at most a^{-4} and eq. 8.xii bounds the torus kernel by Cr_z^{-4}. Here $1 \leq r_z/a$ is bounded above by the fixed threshold. Increasing C gives the same claimed raw-displacement bound.

The shift $a(c_P - c_Q)$ from origins to centres has length at most a. When $r \geq 10a$, the shifted segment has torus distance at least $r-a$ from zero. The first derivative bound in eq. 8.xii gives an error of at most Car^{-5} for that shift. This proves eq. 8.x. Applying the second estimate in eq. 8.xii at u proves eq. 8.xi. □

8.3. **Separated tests and a coarsened Gaussian.**

Proposition 8.4 (Limits of the separated lattice cumulants)**.** For $n \geq 2$ and continuous tests on the torus with pairwise separated supports, the mixed cumulant of $Q_a(f_i)$ converges, as $a \downarrow 0$ through subdivisions of ℓ, to the integral of eq. 8.ii with K_ℓ in place of K. For periodizations of tests in a fixed compact subset of $\mathbb{R}^4$ with positive pairwise separation, it converges to the integral of eq. 8.ii along every sequence with $a \downarrow 0$, $\ell \to \infty$ and ℓ/a integral. The same Euclidean integral is obtained by taking the fixed-volume limit first and then increasing ℓ.

For a quantitative version, let n be fixed and let the tests be real Lipschitz functions, each supported in a ball of radius d in a common Euclidean lift. Suppose every distance between a point of one support and a point of a distinct support lies in $[d, C_{\mathrm{sh}}d]$, $10a \leq d$ and $C_{\mathrm{sh}}d \leq \ell/10$, and that $\mathrm{Lip}(f_i) \leq M\|f_i\|_\infty/d$. The difference between the lattice cumulant and the integral of eq. 8.ii is bounded by

$$C_{n,M,C_{\mathrm{sh}}} d_G g^{2n} \prod_i \|f_i\|_\infty [(a/d)^{1/2} + (d/\ell)^4]. \tag{8.xiii}$$

For odd $n \geq 3$ this is an absolute bound on the lattice cumulant. ■

Proof. Use the finite cycle sum following eq. 8.vii. Each edge of a contributing cycle joins distinct supports. The convergence of every kernel entry is uniform there by Proposition 8.3. The plaquette counts in the sum, after multiplication by a^{4n}, are bounded on a fixed compact support stock. For each set of plane labels the limit is the Riemann integral on the corresponding shifted grids. The integrand is continuous after being assigned the value zero in a neighborhood of each partial diagonal, since the product of the tests vanishes there. Summing the plane labels gives the trace in the cycle. The fixed-volume and joint limits follow from the two kernel estimates. The periodic-to-Euclidean difference after taking the fixed-volume limit tends to zero by eq. 8.xii.

For the rate, write the difference of the cycle products as a sum in which one kernel changes in each term. All kernels are $O(d^{-4})$ on the contributing configurations. By eq. 8.xi the changed kernel has the additional factor $C[(a/d)^{1/2} + (C_{\mathrm{sh}}d/\ell)^4]$. The a^{4n}-weighted count of sites on all test supports is $O(d^{4n})$. This bounds the lattice kernel error by the right side of eq. 8.xiii. The Riemann error of the Euclidean integrand has the additional factor $C_{n,M}a/d$ in place of the bracket. Indeed each kernel has first derivative $O(d^{-5})$ on the cells of the contributing stock, enlarged by one mesh, and the same product estimate uses the stated Lipschitz bounds of the tests. The total volume of these cells is $O(d^{4n})$. Since $a/d \leq 1$, this last error is absorbed in the displayed bound. The odd case uses the exact continuum cancellation in Proposition 8.2. □

Corollary 8.5 (Finite volume for continuous separated tests)**.** Let f, h be nonzero nonnegative continuous compactly supported tests in a common Euclidean lift. Suppose all cross-distances between their supports lie in $[d, C_{\mathrm{sh}}d]$, where $d > 0$, $C_{\mathrm{sh}} \geq 1$, and $C_{\mathrm{sh}}d \leq \ell/4$. Denote the limiting torus two-point covariance of their periodizations, as in proposition 8.4, by $\mathcal{C}_\ell(f, h)$. Then

$$\left| \frac{\mathcal{C}_\ell(f,h)}{C_G g^4 \mathcal{K}(f,h)} - 1 \right| \leq C_{C_{\mathrm{sh}}} (d/\ell)^4.$$

No Lipschitz or lower-mass assumption is required for this finite-volume comparison of the limiting kernels. ■□

Proof. The two-point Wick formula gives

$$\mathcal{C}_\ell(f,h) = \frac{d_G g^4}{2} \iint f(x)h(y) \|\mathsf{K}_\ell(x-y)\|_{\mathrm{HS}}^2 \, dx \, dy.$$

Equation 8.xii and $\|\mathsf{K}(z)\|_{\mathrm{HS}} \leq C|z|^{-4}$ give

$$\left| \|\mathsf{K}_\ell(z)\|_{\mathrm{HS}}^2 - \|\mathsf{K}(z)\|_{\mathrm{HS}}^2 \right| \leq C\ell^{-4}|z|^{-4} \quad (0 < |z| \leq \ell/4).$$

Multiply by $f(x)h(y) \geq 0$ and use $|x-y|^{-4} \leq (C_{\mathrm{sh}}d)^4 |x-y|^{-8}$. The denominator is positive because both tests are continuous, nonnegative and nonzero. Division proves the bound. □

The odd-cycle cancellation pertains to the Euclidean limit. For example, group the six plaquettes at each raw base site on the torus with $a = 1, \ell = 2$. For one colour at unit coupling, the third cumulant of their centred half-squares at the three base sites $0, e_4, e_3$ equals $25/36864$. To evaluate it directly, let $A(z)$ be the matrix in eq. 8.ix before taking its PQ entry. If $b_S(v) = \mathbf{1}_S \wedge v$, with $\mathbf{1}_S$ the coordinate indicator vector of $S \subset \{1,2,3,4\}$, the two-site frequencies give

$$192A(z) = \sum_{S \neq \varnothing} \frac{12}{|S|} (-1)^{\sum_{\mu \in S} z_\mu} b_S b_S^{\mathsf{T}}.$$

In the ordered plane basis $12, 13, 14, 23, 24, 34$ the diagonal of $(192A(e_4))(192A(e_3+e_4))(192A(e_3))$ is $(20096, 6240, 6240, 6240, 6240, -40256)$. All six ordered cycles in the third cumulant have the same trace, by cyclicity and transpose. Dividing the sum of these diagonal entries by 192^3 gives the stated cumulant.

For the two-point estimate fix $d > 0$ and $0 < \vartheta \leq 1$. Take nonzero, nonnegative Lipschitz tests f, h, each supported in a ball of radius d, with

$$\begin{gathered} d \leq \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \leq 6d, \\ \int f \geq \vartheta \|f\|_\infty d^4, \qquad \operatorname{Lip}(f) \leq \|f\|_\infty/(\vartheta d), \end{gathered} \tag{8.xiv}$$

and the same last two inequalities for h. Amplitudes are arbitrary. Dividing by the amplitude and dilating a support ball to radius one gives a profile with integral at least ϑ and Lipschitz constant at most ϑ^{-1}. For $d \leq \ell/100$ these conditions also define a class on the torus. To evaluate $\mathcal{K}$ there, lift the two support balls so a minimizing pair of support points has the short displacement. Each ball lifts injectively, and all resulting cross-displacements have length at most $10d$ by the two support diameters. Their Euclidean lengths are the torus distances. The value of $\mathcal{K}$ is thus independent of the simultaneous periodic translate of the chosen lifts.

To describe coarsening let L be the fixed odd block factor and $b = L^j a$ for a nonnegative integer j. For $j \geq 1$ use the compatible periodic hierarchy on which each period contains an integer number of blocks at scale b and at least two coarse sites. For the corresponding projections P_j, Proposition 3.2 identifies the conditional mean curvature as the Gaussian with covariance $g^2 P_j$. Let $Q_a^{[j]}$ be eq. 8.vi for that curvature, centred with respect to its own law.

Theorem D (The uniform separated two-point comparison)**.** For the shape class eq. 8.xiv with $10a \le d \le \ell/100$, the full curvature covariance obeys

$$\left|\frac{\mathrm{Cov}(Q_a(f),Q_a(h))}{C_G g^4 \mathcal{K}(f,h)} - 1\right| \le C\big[(a/d)^{1/2} + (d/\ell)^4 + a/(\vartheta^3 d)\big].$$

The denominator is strictly positive. If $b = L^j a \le d$ in the compatible hierarchy, then the corresponding bound for $Q_a^{[j]}$ is

$$\left|\frac{\mathrm{Cov}(Q_a^{[j]}(f),Q_a^{[j]}(h))}{C_G g^4 \mathcal{K}(f,h)} - 1\right| \le C_L\big[(a/d)^{1/2} + (d/\ell)^4 + a/(\vartheta^3 d) + \mathbf{1}_{\{j\ge 1\}} e^{-c_L d/b}\big]. \tag{8.xv}$$

The constants are independent of depth, the torus side, mesh, test amplitudes and positions, and g. ■□

Proof. The supports and mass lower bounds give

$$c\vartheta^2 \|f\|_\infty \|h\|_\infty \le \mathcal{K}(f,h) \le C\|f\|_\infty \|h\|_\infty.$$

For the upper bound use the volume of the two balls and the lower separation. For the lower bound use the maximum distance $10d$ and the two mass bounds.

For $j \ge 1$, [2, Proposition 6.4] gives, for $1 \le i \le j$, the counting-coordinate entry bound

$$|(P_{i-1} - P_i)(p,q)| \le C_L (a/b_i)^4 e^{-c_L r/b_i}, \qquad b_i = L^{i-1}a.$$

Here r may be taken as the barycentre distance, because moving the chosen plaquette reference sites by $O(a)$ changes the exponential by a fixed factor. Multiplying by a^{-4} and summing gives a physical kernel error bounded by $C_L r^{-4} e^{-c_L' r/b}$ for $r \ge b$. In fact $r/b_i = L^{j-i+1} r/b$, and the sum of $(r/b_i)^4 e^{-c_L r/b_i}$ is bounded by a smaller exponential in r/b. For $j = 0$ the difference vanishes. This derives the coarsened kernel error by subtracting the fluctuation projections from the full projection.

On the cross-support lattice pairs write E for the sum of the mesh and volume errors and, for the coarsened law, the exponential error on the right of eq. 8.xv, without the quadrature term. Each physical entry of the relevant covariance divided by g^2 differs from $\mathsf{K}_{PQ}(u)$ by at most $C_L E|u|^{-4}$. Its squared entry therefore differs by at most $C_L E|u|^{-8}$, since E is bounded in the stated range. Positivity of the tests bounds the summed error by this factor times the corresponding Riemann sum of $f(x)h(y)|u|^{-8}$.

For each fixed pair of plane labels, the shifted-grid quadrature for $f(x)h(y)|u|^{-8}$ has absolute error at most $Ca\|f\|_\infty\|h\|_\infty/(\vartheta d)$. Indeed the mesh cells meeting the product of the supports are contained in the product of two balls of radius $d + 2a$, and throughout them the cross-distance is bounded below by $d - 4a$. The product rule bounds the Lipschitz norm of the integrand there by $C\vartheta^{-1} d^{-9}\|f\|_\infty\|h\|_\infty$. Using the cell diameter Ca and the product volume Cd^8 proves the bound. The same argument applies to the integrand with $\mathsf{K}_{PQ}(u)^2$ in place of $|u|^{-8}$, by Lemma 8.1 and its differentiated expression. The relative quadrature cost is at most $Ca/(\vartheta^3 d)$ by the lower bound on $\mathcal{K}$. Multiplication of that cost by the bounded E still gives a term of the stated order.

Finally sum over the planes and apply Lemma 8.1 to the continuum integral. The Wick factor $d_G g^4/2$ in eq. 8.vii then gives the coefficient in eq. 8.iii. □

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